CHAPTER

14 Operational Amplifiers

14.1 INTRODUCTION

An operational amplifier, or op-amp, is a very high gain differential amplifier with high input impedance and low output impedance. Typical uses of the operational amplifier are to provide voltage amplitude changes (amplitude and polarity), oscillators, filter circuits, and many types of instrumentation circuits. An op-amp contains a number of differential amplifier stages to achieve a very high voltage gain.

Figure 14.1 shows a basic op-amp with two inputs and one output as would result using a differential amplifier input stage. Recall from Chapter 12 that each input results in either the same or an opposite polarity (or phase) output, depending on whether the signal is applied to the plus (+) or the minus (−) input.

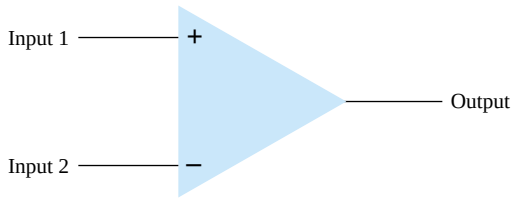


Figure 14.1 Basic op-amp.

Single-Ended Input

Single-ended input operation results when the input signal is connected to one input with the other input connected to ground. Figure 14.2 shows the signals connected

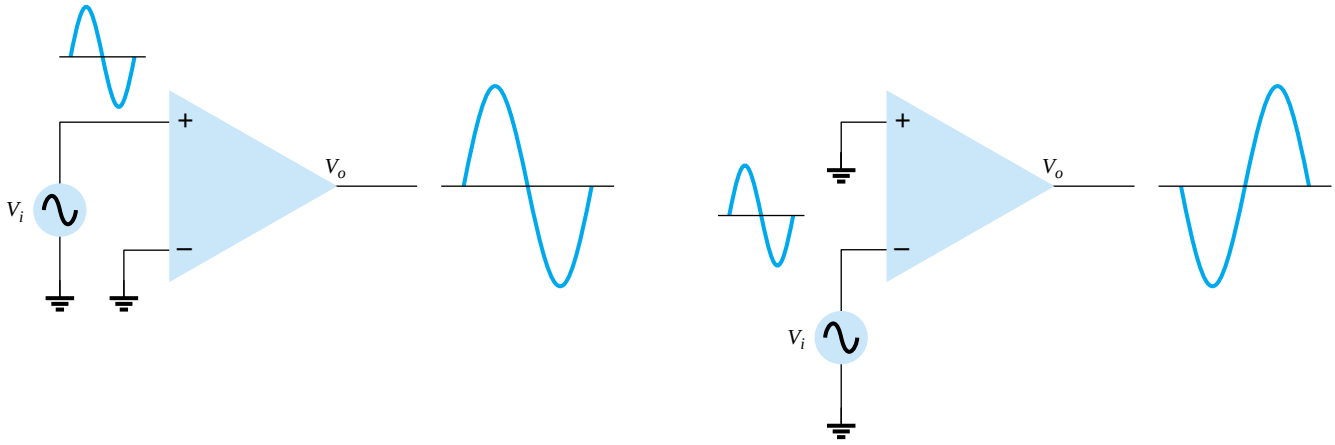
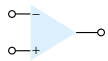


Figure 14.2 Single-ended operation.



for this operation. In Fig. 14.2a, the input is applied to the plus input (with minus input at ground), which results in an output having the same polarity as the applied input signal. Figure 14.2b shows an input signal applied to the minus input, the output then being opposite in phase to the applied signal.

Double-Ended (Differential) Input

In addition to using only one input, it is possible to apply signals at each input—this being a double-ended operation. Figure 14.3a shows an input, V_d, applied between the two input terminals (recall that neither input is at ground), with the resulting amplified output in phase with that applied between the plus and minus inputs. Figure 14.3b shows the same action resulting when two separate signals are applied to the inputs, the difference signal being $V_{i_1} - V_{i_2}$.

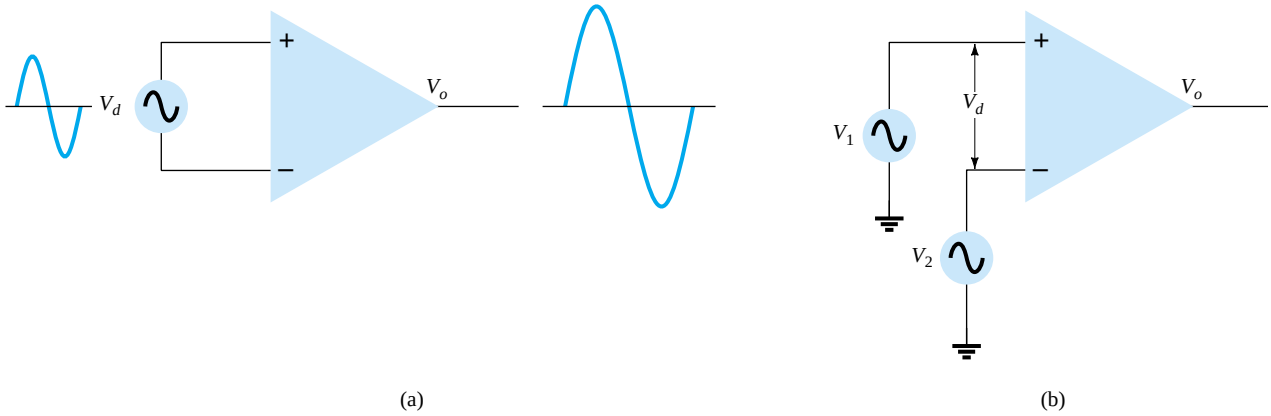


Figure 14.3 Double-ended (differential) operation.

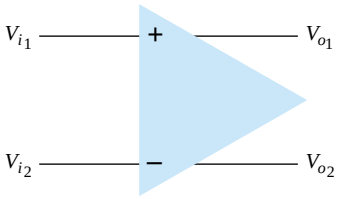


Figure 14.4 Double-ended output.

Double-Ended Output

While the operation discussed so far had a single output, the op-amp can also be operated with opposite outputs, as shown in Fig. 14.4. An input applied to either input will result in outputs from both output terminals, these outputs always being opposite in polarity. Figure 14.5 shows a single-ended input with a double-ended output. As shown, the signal applied to the plus input results in two amplified outputs of opposite polarity. Figure 14.6 shows the same operation with a single output measured

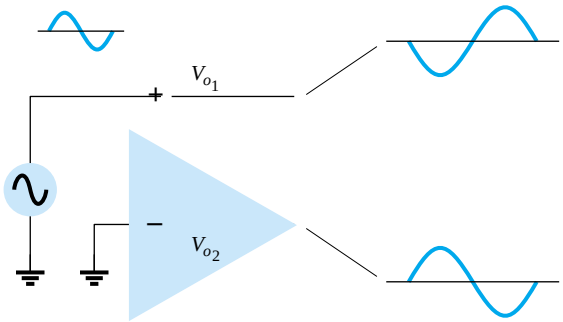


Figure 14.5 Double-ended output with single-ended input.

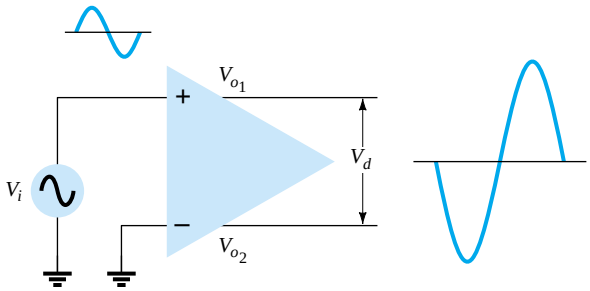


Figure 14.6 Double-ended output.

between output terminals (not with respect to ground). This difference output signal is $V_{o_1} - V_{o_2}$. The difference output is also referred to as a *floating signal* since neither output terminal is the ground (reference) terminal. Notice that the difference output is twice as large as either V_{o_1} or V_{o_2} since they are of opposite polarity and subtracting them results in twice their amplitude [i.e., 10 V − (−10 V) = 20 V]. Figure 14.7 shows a differential input, differential output operation. The input is applied between the two input terminals and the output taken from between the two output terminals. This is fully differential operation.

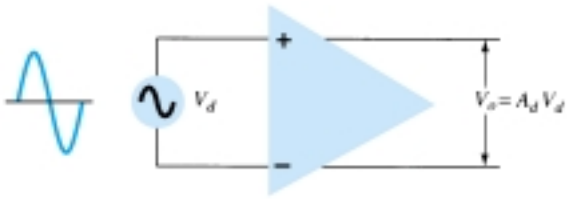


Figure 14.7 Differential-input, differential-output operation.

Common-Mode Operation

When the same input signals are applied to both inputs, common-mode operation results, as shown in Fig. 14.8. Ideally, the two inputs are equally amplified, and since they result in opposite polarity signals at the output, these signals cancel, resulting in 0-V output. Practically, a small output signal will result.

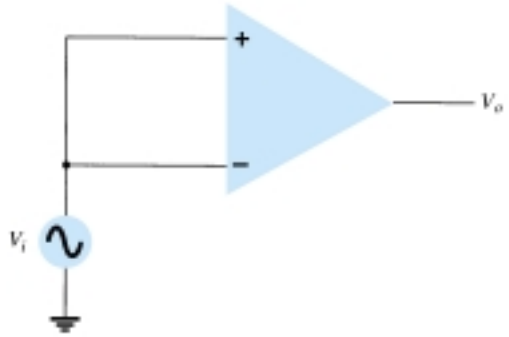


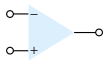
Figure 14.8 Common-mode operation.

Common-Mode Rejection

A significant feature of a differential connection is that the signals which are opposite at the inputs are highly amplified, while those which are common to the two inputs are only slightly amplified—the overall operation being to amplify the difference signal while rejecting the common signal at the two inputs. Since noise (any unwanted input signal) is generally common to both inputs, the differential connection tends to provide attenuation of this unwanted input while providing an amplified output of the difference signal applied to the inputs. This operating feature, referred to as common-mode rejection, is discussed more fully in the next section.

14.2 DIFFERENTIAL AND COMMON-MODE OPERATION

One of the more important features of a differential circuit connection, as provided in an op-amp, is the circuit's ability to greatly amplify signals that are opposite at the two inputs, while only slightly amplifying signals that are common to both inputs. An



op-amp provides an output component that is due to the amplification of the difference of the signals applied to the plus and minus inputs and a component due to the signals common to both inputs. Since amplification of the opposite input signals is much greater than that of the common input signals, the circuit provides a common-mode rejection as described by a numerical value called the common-mode rejection ratio (CMRR).

Differential Inputs

When separate inputs are applied to the op-amp, the resulting difference signal is the difference between the two inputs.

$$V_d = V_{i_1} - V_{i_2} \tag{14.1}$$

Common Inputs

When both input signals are the same, a common signal element due to the two inputs can be defined as the average of the sum of the two signals.

$$V_c = \tfrac{1}{2}(V_{i_1} + V_{i_2}) \tag{14.2}$$

Output Voltage

Since any signals applied to an op-amp in general have both in-phase and out-of-phase components, the resulting output can be expressed as

$$V_o = A_d V_d + A_c V_c \tag{14.3}$$

where V_d = difference voltage given by Eq. (14.1)
V_c = common voltage given by Eq. (14.2)
A_d = differential gain of the amplifier
A_c = common-mode gain of the amplifier

Opposite Polarity Inputs

If opposite polarity inputs applied to an op-amp are ideally opposite signals, $V_{i_1} = -V_{i_2} = V_s$, the resulting difference voltage is

$$\text{Eq. (14.1): } V_d = V_{i_1} - V_{i_2} = V_s - (-V_s) = 2V_s$$

while the resulting common voltage is

$$\text{Eq. (14.2): } V_c = \tfrac{1}{2}(V_{i_1} + V_{i_2}) = \tfrac{1}{2}[V_s + (-V_s)] = 0$$

so that the resulting output voltage is

$$\text{Eq. (14.3): } V_o = A_d V_d + A_c V_c = A_d\,(2V_s) + 0 = 2\,A_d V_s$$

This shows that when the inputs are an ideal opposite signal (no common element), the output is the differential gain times twice the input signal applied to one of the inputs.

Same Polarity Inputs

If the same polarity inputs are applied to an op-amp, $V_{i_1} = V_{i_2} = V_s$, the resulting difference voltage is

$$\text{Eq. (14.1): } V_d = V_{i_1} - V_{i_2} = V_s - V_s = 0$$

while the resulting common voltage is

$$\text{Eq. (14.2):} \quad V_c = \tfrac{1}{2}(V_{i_1} + V_{i_2}) = \tfrac{1}{2}(V_s + V_s) = V_s$$

so that the resulting output voltage is

$$\text{Eq. (14.3):} \quad V_o = A_d V_d + A_c V_c = A_d(0) + A_c V_s = A_c V_s$$

This shows that when the inputs are ideal in-phase signals (no difference signal), the output is the common-mode gain times the input signal, V_s, which shows that only common-mode operation occurs.

Common-Mode Rejection

The solutions above provide the relationships that can be used to measure A_d and A_c in op-amp circuits.

1. *To measure A_d:* Set $V_{i_1} = -V_{i_2} = V_s = 0.5$ V, so that

   $$\text{Eq. (14.1):} \quad V_d = (V_{i_1} - V_{i_2}) = (0.5\text{ V} - (-0.5\text{ V}) = 1\text{ V}$$

   and $$\text{Eq. (14.2):} \quad V_c = \tfrac{1}{2}(V_{i_1} + V_{i_2}) = \tfrac{1}{2}[0.5\text{ V} + (-0.5\text{ V})] = 0\text{ V}$$

   Under these conditions the output voltage is

   $$\text{Eq. (14.3):} \quad V_o = A_d V_d + A_c V_c = A_d(1\text{ V}) + A_c(0) = A_d$$

   Thus, setting the input voltages $V_{i_1} = -V_{i_2} = 0.5$ V results in an output voltage numerically equal to the value of A_d.
2. *To measure A_c:* Set $V_{i_1} = V_{i_2} = V_s = 1$ V, so that

   $$\text{Eq. (14.1):} \quad V_d = (V_{i_1} - V_{i_2}) = (1\text{ V} - 1\text{ V}) = 0\text{ V}$$

   and $$\text{Eq. (14.2):} \quad V_c = \tfrac{1}{2}(V_{i_1} + V_{i_2}) = \tfrac{1}{2}(1\text{ V} + 1\text{ V}) = 1\text{ V}$$

   Under these conditions the output voltage is

   $$\text{Eq. (14.3):} \quad V_o = A_d V_d + A_c V_c = A_d(0\text{ V}) + A_c(1\text{ V}) = A_c$$

   Thus, setting the input voltages $V_{i_1} = V_{i_2} = 1$ V results in an output voltage numerically equal to the value of A_c.

Common-Mode Rejection Ratio

Having obtained A_d and A_c (as in the measurement procedure discussed above), we can now calculate a value for the common-mode rejection ratio (CMRR), which is defined by the following equation:

$$\text{CMRR} = \frac{A_d}{A_c} \tag{14.4}$$

The value of CMRR can also be expressed in logarithmic terms as

$$\text{CMRR (log)} = 20 \log_{10} \frac{A_d}{A_c} \quad \text{(dB)} \tag{14.5}$$

EXAMPLE 14.1

Calculate the CMRR for the circuit measurements shown in Fig. 14.9.

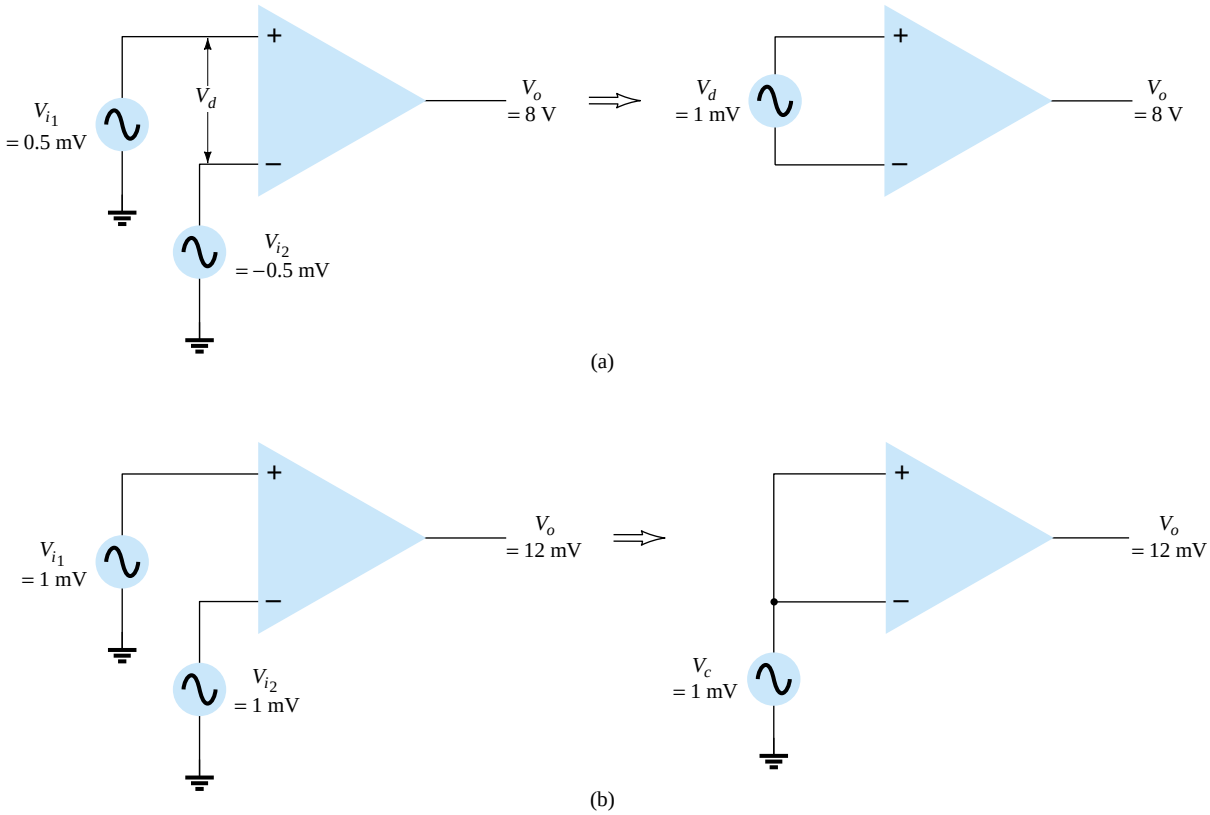


Figure 14.9 Differential and common-mode operation: (a) differential-mode; (b) common-mode.

Solution

From the measurement shown in Fig. 14.9a, using the procedure in step 1 above, we obtain

$$A_d = \frac{V_o}{V_d} = \frac{8 \text{ V}}{1 \text{ mV}} = 8000$$

The measurement shown in Fig. 14.9b, using the procedure in step 2 above, gives us

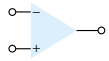
$$A_c = \frac{V_o}{V_c} = \frac{12 \text{ mV}}{1 \text{ mV}} = 12$$

Using Eq. (14.4), the value of CMRR is

$$\text{CMRR} = \frac{A_d}{A_c} = \frac{8000}{12} = \mathbf{666.7}$$

which can also be expressed as

$$\text{CMRR} = 20 \log_{10} \frac{A_d}{A_c} = 20 \log_{10} 666.7 = \mathbf{56.48 \text{ dB}}$$



It should be clear that the desired operation will have A_d very large with A_c very small. That is, the signal components of opposite polarity will appear greatly amplified at the output, whereas the signal components that are in phase will mostly cancel out so that the common-mode gain, A_c, is very small. Ideally, the value of the CMRR is infinite. Practically, the larger the value of CMRR, the better the circuit operation.

We can express the output voltage in terms of the value of CMRR as follows:

$$\text{Eq. (14.3):} \quad V_o = A_d V_d + A_c V_c = A_d V_d \left(1 + \frac{A_c V_c}{A_d V_d}\right)$$

Using Eq. (14.4), we can write the above as

$$V_o = A_d V_d \left(1 + \frac{1}{\text{CMRR}} \frac{V_c}{V_d}\right) \tag{14.6}$$

Even when both V_d and V_c components of signal are present, Eq. (14.6) shows that for large values of CMRR, the output voltage will be due mostly to the difference signal, with the common-mode component greatly reduced or rejected. Some practical examples should help clarify this idea.

EXAMPLE 14.2

Determine the output voltage of an op-amp for input voltages of $V_{i_1} = 150\ \mu\text{V}$, $V_{i_2} = 140\ \mu\text{V}$. The amplifier has a differential gain of $A_d = 4000$ and the value of CMRR is:

(a) 100.

(b) 10^5.

Solution

$$\text{Eq. (14.1):} \quad V_d = V_{i_1} - V_{i_2} = (150 - 140)\ \mu\text{V} = 10\ \mu\text{V}$$

$$\text{Eq. (14.2):} \quad V_c = \frac{1}{2}(V_{i_1} + V_{i_2}) = \frac{150\ \mu\text{V} + 140\ \mu\text{V}}{2} = 145\ \mu\text{V}$$

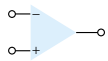
(a) Eq. (14.6):
$$V_o = A_d V_d \left(1 + \frac{1}{\text{CMRR}} \frac{V_c}{V_d}\right)$$
$$= (4000)(10\ \mu\text{V})\left(1 + \frac{1}{100} \frac{145\ \mu\text{V}}{10\ \mu\text{V}}\right)$$
$$= 40\ \text{mV}(1.145) = \mathbf{45.8\ mV}$$

(b) $V_o = (4000)(10\ \mu\text{V})\left(1 + \frac{1}{10^5} \frac{145\ \mu\text{V}}{10\ \mu\text{V}}\right) = 40\ \text{mV}(1.000145) = \mathbf{40.006\ mV}$

Example 14.2 shows that the larger the value of CMRR, the closer the output voltage is to the difference input times the difference gain with the common-mode signal being rejected.

14.3 OP-AMP BASICS

An operational amplifier is a very high gain amplifier having very high input impedance (typically a few megohms) and low output impedance (less than 100 Ω). The basic circuit is made using a difference amplifier having two inputs (plus and minus) and at least one output. Figure 14.10 shows a basic op-amp unit. As discussed ear-



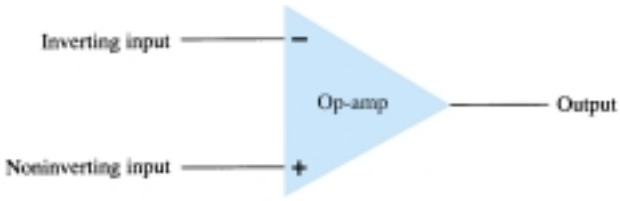


Figure 14.10 Basic op-amp.

lier, the plus (+) input produces an output that is in phase with the signal applied, while an input to the minus (−) input results in an opposite polarity output. The ac equivalent circuit of the op-amp is shown in Fig. 14.11a. As shown, the input signal applied between input terminals sees an input impedance, R_i, typically very high. The output voltage is shown to be the amplifier gain times the input signal taken through an output impedance, R_o, which is typically very low. An ideal op-amp circuit, as shown in Fig. 14.11b, would have infinite input impedance, zero output impedance, and an infinite voltage gain.

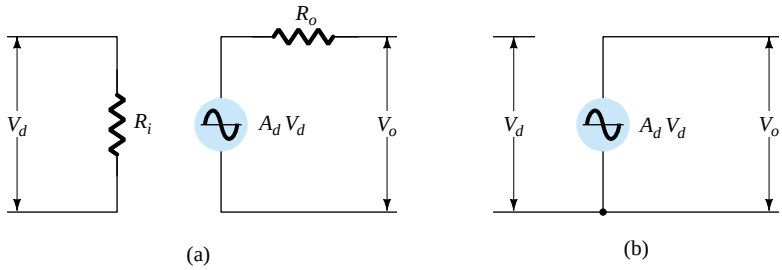


Figure 14.11 Ac equivalent of op-amp circuit: (a) practical; (b) ideal.

Basic Op-Amp

The basic circuit connection using an op-amp is shown in Fig. 14.12. The circuit shown provides operation as a constant-gain multiplier. An input signal, V_1, is applied through resistor R_1 to the minus input. The output is then connected back to the same minus input through resistor R_f. The plus input is connected to ground. Since the signal V_1 is essentially applied to the minus input, the resulting output is opposite in phase to the input signal. Figure 14.13a shows the op-amp replaced by its ac equivalent circuit. If we use the ideal op-amp equivalent circuit, replacing R_i by an infinite resistance and R_o by zero resistance, the ac equivalent circuit is that shown in Fig. 14.13b. The circuit is then redrawn, as shown in Fig. 14.13c, from which circuit analysis is carried out.

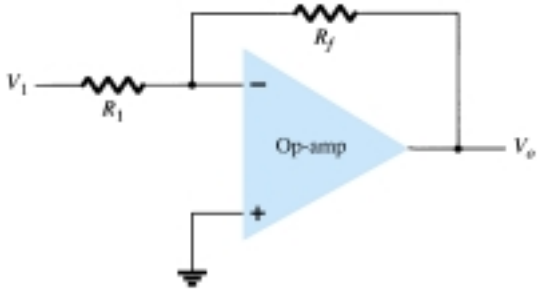


Figure 14.12 Basic op-amp connection.

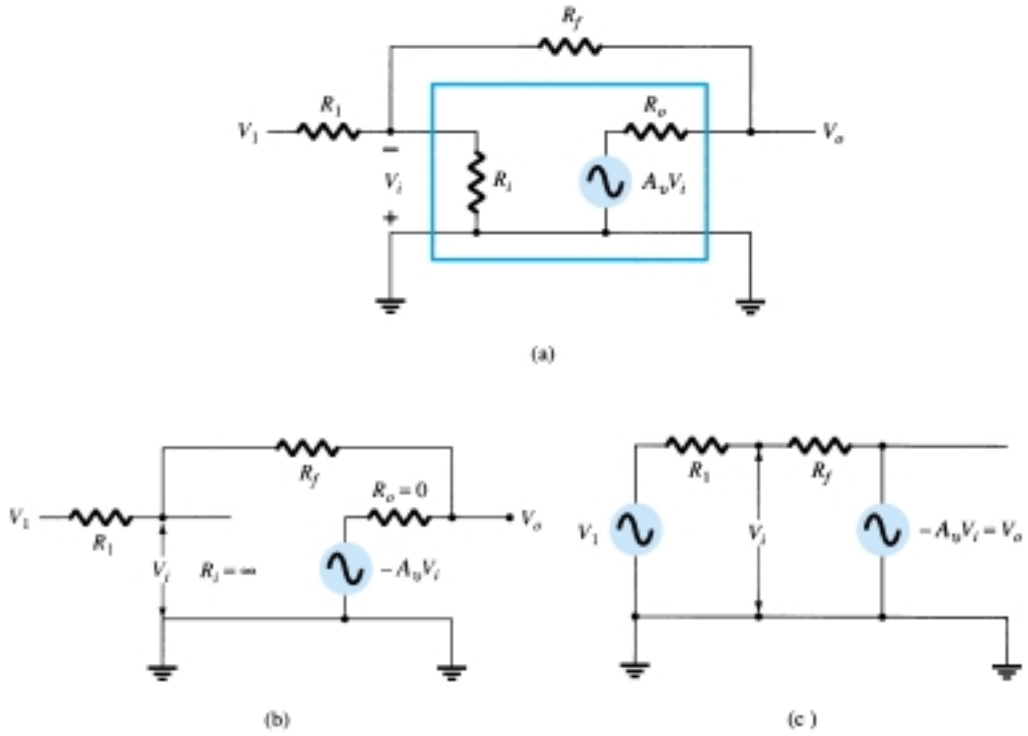


Figure 14.13 Operation of op-amp as constant-gain multiplier: (a) op-amp ac equivalent circuit; (b) ideal op-amp equivalent circuit; (c) redrawn equivalent circuit.

Using superposition, we can solve for the voltage V_1 in terms of the components due to each of the sources. For source V_1 only ($-A_vV_i$ set to zero),

$$V_{i_1} = \frac{R_f}{R_1 + R_f} V_1$$

For source $-A_vV_i$ only (V_1 set to zero),

$$V_{i_2} = \frac{R_1}{R_1 + R_f} (-A_vV_i)$$

The total voltage V_i is then

$$V_i = V_{i_1} + V_{i_2} = \frac{R_f}{R_1 + R_f} V_1 + \frac{R_1}{R_1 + R_f} (-A_vV_i)$$

which can be solved for V_i as

$$V_i = \frac{R_f}{R_f + (1 + A_v)R_1} V_1 \tag{14.7}$$

If $A_v \gg 1$ and $A_v R_1 \gg R_f$, as is usually true, then

$$V_i = \frac{R_f}{A_vR_1} V_1$$

Solving for V_o/V_i, we get

$$\frac{V_o}{V_i} = \frac{-A_vV_i}{V_i} = \frac{-A_v}{V_i} \frac{R_fV_1}{A_vR_1} = -\frac{R_f}{R_1} \frac{V_1}{V_i}$$

so that

$$\frac{V_o}{V_1} = -\frac{R_f}{R_1} \tag{14.8}$$

The result, in Eq. (14.8), shows that the ratio of overall output to input voltage is dependent only on the values of resistors R_1 and R_f —provided that A_v is very large.

Unity Gain

If $R_f = R_1$, the gain is

$$\text{voltage gain} = -\frac{R_f}{R_1} = -1$$

so that the circuit provides a unity voltage gain with 180° phase inversion. If R_f is exactly R_1, the voltage gain is exactly 1.

Constant Magnitude Gain

If R_f is some multiple of R_1, the overall amplifier gain is a constant. For example, if $R_f = 10R_1$, then

$$\text{voltage gain} = -\frac{R_f}{R_1} = -10$$

and the circuit provides a voltage gain of exactly 10 along with an 180° phase inversion from the input signal. If we select precise resistor values for R_f and R_1, we can obtain a wide range of gains, the gain being as accurate as the resistors used and is only slightly affected by temperature and other circuit factors.

Virtual Ground

The output voltage is limited by the supply voltage of, typically, a few volts. As stated before, voltage gains are very high. If, for example, $V_o = -10$ V and $A_v = 20{,}000$, the input voltage would then be

$$V_i = \frac{-V_o}{A_v} = \frac{10 \text{ V}}{20{,}000} = 0.5 \text{ mV}$$

If the circuit has an overall gain (V_o/V_1) of, say, 1, the value of V_1 would then be 10 V. Compared to all other input and output voltages, the value of V_i is then small and may be considered 0 V.

Note that although $V_i \approx 0$ V, it is not exactly 0 V. (The output voltage is a few volts due to the very small input V_i times a very large gain A_v.) The fact that $V_i \approx 0$ V leads to the concept that at the amplifier input there exists a virtual short circuit or virtual ground.

The concept of a virtual short implies that although the voltage is nearly 0 V, there is no current through the amplifier input to ground. Figure 14.14 depicts the virtual ground concept. The heavy line is used to indicate that we may consider that a short

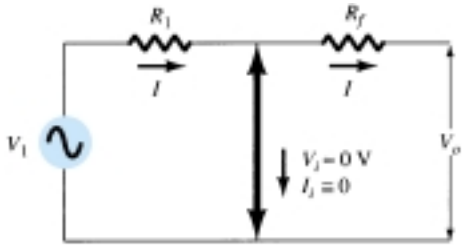


Figure 14.14 Virtual ground in an op-amp.

exists with $V_i \approx 0$ V but that this is a virtual short so that no current goes through the short to ground. Current goes only through resistors R_1 and R_f as shown.

Using the virtual ground concept, we can write equations for the current I as follows:

$$I = \frac{V_1}{R_1} = -\frac{V_o}{R_f}$$

which can be solved for V_o/V_1:

$$\frac{V_o}{V_1} = -\frac{R_f}{R_1}$$

The virtual ground concept, which depends on A_v being very large, allowed a simple solution to determine the overall voltage gain. It should be understood that although the circuit of Fig. 14.14 is not physically correct, it does allow an easy means for determining the overall voltage gain.

14.4 PRACTICAL OP-AMP CIRCUITS

The op-amp can be connected in a large number of circuits to provide various operating characteristics. In this section, we cover a few of the most common of these circuit connections.

Inverting Amplifier

The most widely used constant-gain amplifier circuit is the inverting amplifier, as shown in Fig. 14.15. The output is obtained by multiplying the input by a fixed or constant gain, set by the input resistor (R_1) and feedback resistor (R_f)—this output also being inverted from the input. Using Eq. (14.8) we can write

$$V_o = -\frac{R_f}{R_1} V_1$$

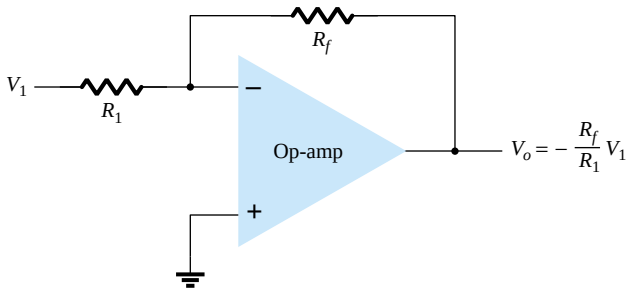


Figure 14.15 Inverting constant-gain multiplier.

EXAMPLE 14.3

If the circuit of Fig. 14.15 has $R_1 = 100$ kΩ and $R_f = 500$ kΩ, what output voltage results for an input of $V_1 = 2$ V?

Solution

$$\text{Eq. (14.8): } V_o = -\frac{R_f}{R_1} V_1 = -\frac{500\ \text{k}\Omega}{100\ \text{k}\Omega} (2\ \text{V}) = \mathbf{-10\ V}$$

Noninverting Amplifier

The connection of Fig. 14.16a shows an op-amp circuit that works as a noninverting amplifier or constant-gain multiplier. It should be noted that the inverting amplifier connection is more widely used because it has better frequency stability (discussed later). To determine the voltage gain of the circuit, we can use the equivalent representation shown in Fig. 14.16b. Note that the voltage across R_1 is V_1 since $V_i \approx 0$ V. This must be equal to the output voltage, through a voltage divider of R_1 and R_f, so that

$$V_1 = \frac{R_1}{R_1 + R_f} V_o$$

which results in

$$\boxed{\frac{V_o}{V_1} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_f}{R_1}} \tag{14.9}$$

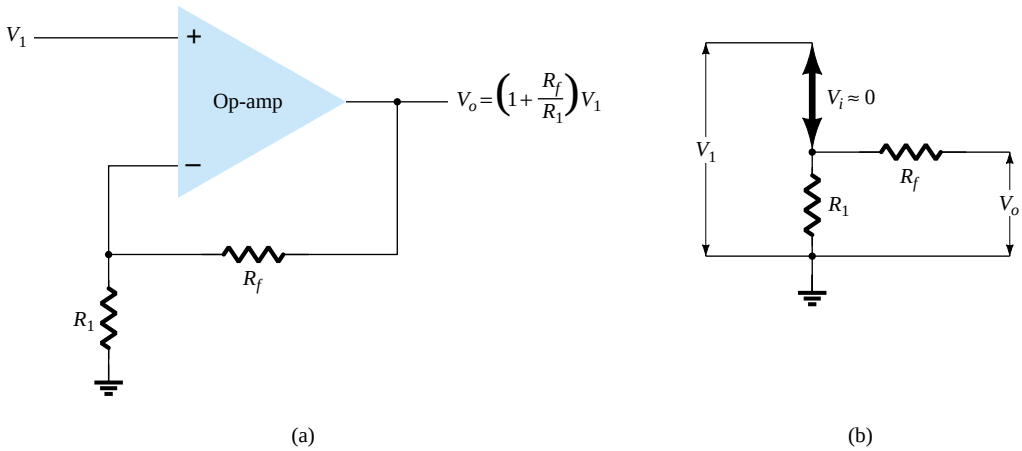


Figure 14.16 Noninverting constant-gain multiplier.

EXAMPLE 14.4

Calculate the output voltage of a noninverting amplifier (as in Fig. 14.16) for values of $V_1 = 2$ V, $R_f = 500$ kΩ, and $R_1 = 100$ kΩ.

Solution

$$\text{Eq. (14.9): } V_o = \left(1 + \frac{R_f}{R_1}\right) V_1 = \left(1 + \frac{500\ \text{k}\Omega}{100\ \text{k}\Omega}\right)(2\ \text{V}) = 6(2\ \text{V}) = \mathbf{+12\ V}$$

Unity Follower

The unity-follower circuit, as shown in Fig. 14.17a, provides a gain of unity (1) with no polarity or phase reversal. From the equivalent circuit (see Fig. 14.17b) it is clear that

$$\boxed{V_o = V_1} \tag{14.10}$$

and that the output is the same polarity and magnitude as the input. The circuit operates like an emitter- or source-follower circuit except that the gain is exactly unity.

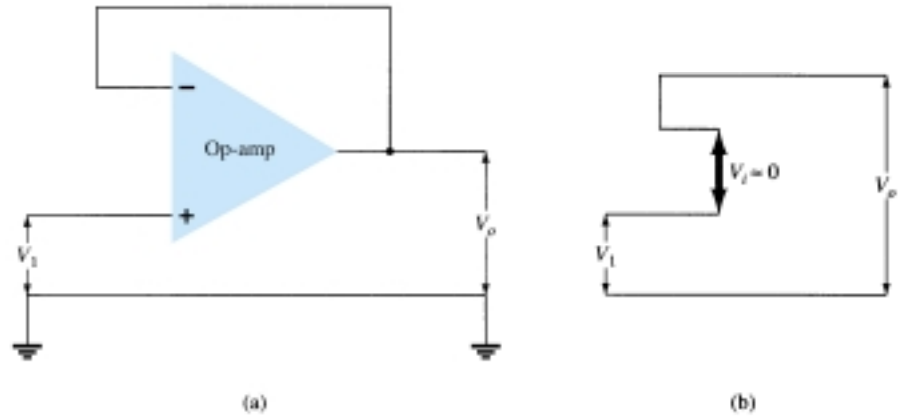


Figure 14.17 (a) Unity follower; (b) virtual-ground equivalent circuit.

Summing Amplifier

Probably the most used of the op-amp circuits is the summing amplifier circuit shown in Fig. 14.18a. The circuit shows a three-input summing amplifier circuit, which provides a means of algebraically summing (adding) three voltages, each multiplied by a constant-gain factor. Using the equivalent representation shown in Fig. 14.18b, the output voltage can be expressed in terms of the inputs as

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right) \tag{14.11}$$

In other words, each input adds a voltage to the output multiplied by its separate constant-gain multiplier. If more inputs are used, they each add an additional component to the output.

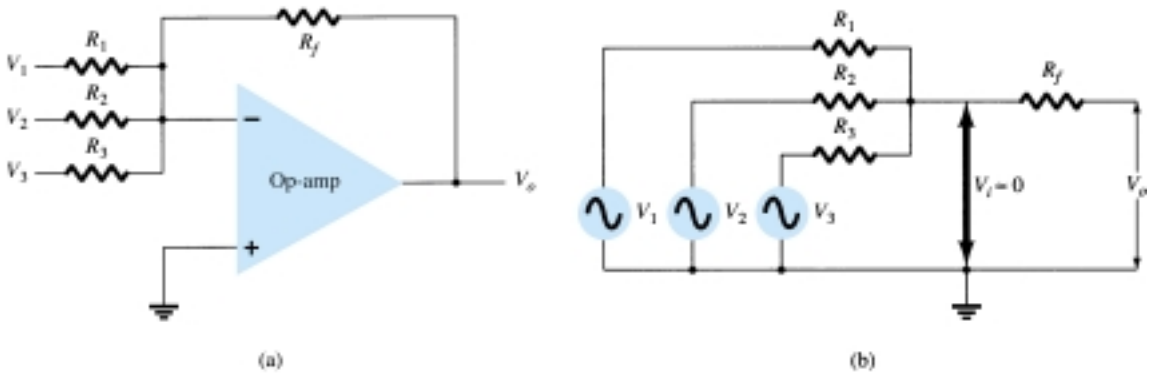


Figure 14.18 (a) Summing amplifier; (b) virtual-ground equivalent circuit.

EXAMPLE 14.5

Calculate the output voltage of an op-amp summing amplifier for the following sets of voltages and resistors. Use $R_f = 1$ MΩ in all cases.

(a) $V_1 = +1$ V, $V_2 = +2$ V, $V_3 = +3$ V, $R_1 = 500$ kΩ, $R_2 = 1$ MΩ, $R_3 = 1$ MΩ.

(b) $V_1 = -2$ V, $V_2 = +3$ V, $V_3 = +1$ V, $R_1 = 200$ kΩ, $R_2 = 500$ kΩ, $R_3 = 1$ MΩ.

Solution

Using Eq. (14.11):

$$\text{(a)}\quad V_o = -\left[\frac{1000\ \text{k}\Omega}{500\ \text{k}\Omega}(+1\ \text{V}) + \frac{1000\ \text{k}\Omega}{1000\ \text{k}\Omega}(+2\ \text{V}) + \frac{1000\ \text{k}\Omega}{1000\ \text{k}\Omega}(+3\ \text{V})\right]$$

$$= -[2(1\ \text{V}) + 1(2\ \text{V}) + 1(3\ \text{V})] = \mathbf{-7\ V}$$

$$\text{(b)}\quad V_o = -\left[\frac{1000\ \text{k}\Omega}{200\ \text{k}\Omega}(-2\ \text{V}) + \frac{1000\ \text{k}\Omega}{500\ \text{k}\Omega}(+3\ \text{V}) + \frac{1000\ \text{k}\Omega}{1000\ \text{k}\Omega}(+1\ \text{V})\right]$$

$$= -[5(-2\ \text{V}) + 2(3\ \text{V}) + 1(1\ \text{V})] = \mathbf{+3\ V}$$

Integrator

So far, the input and feedback components have been resistors. If the feedback component used is a capacitor, as shown in Fig. 14.19a, the resulting connection is called an *integrator.* The virtual-ground equivalent circuit (Fig. 14.19b) shows that an expression for the voltage between input and output can be derived in terms of the current I, from input to output. Recall that virtual ground means that we can consider the voltage at the junction of R and X_C to be ground (since $V_i \approx 0$ V) but that no current goes into ground at that point. The capacitive impedance can be expressed as

$$X_C = \frac{1}{j\omega C} = \frac{1}{sC}$$

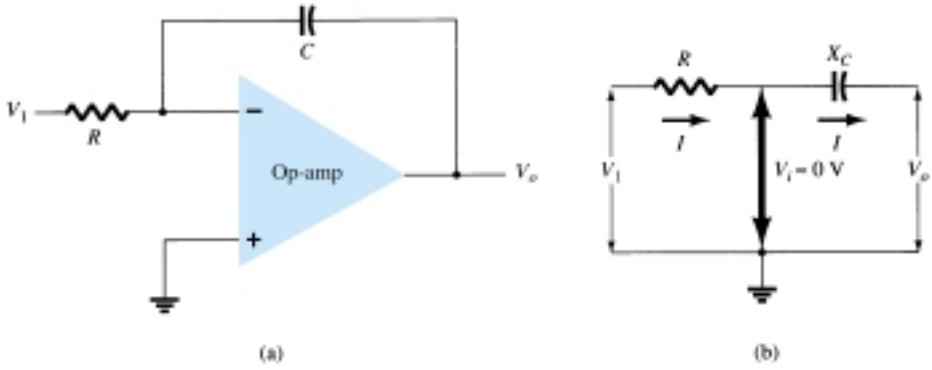


Figure 14.19 Integrator.

where $s = j\omega$ is in the Laplace notation.* Solving for V_o/V_1 yields

$$I = \frac{V_1}{R} = -\frac{V_o}{X_C} = \frac{-V_o}{1/sC} = -sCV_o$$

$$\frac{V_o}{V_1} = \frac{-1}{sCR} \tag{14.12}$$

The expression above can be rewritten in the time domain as

$$\boxed{v_o(t) = -\frac{1}{RC}\int v_1(t)\,dt} \tag{14.13}$$

*Laplace notation allows expressing differential or integral operations which are part of calculus in algebraic form using the operator **s**. Readers unfamiliar with calculus should ignore the steps leading to Eq. (14.13) and follow the physical meaning used thereafter.

Equation (14.13) shows that the output is the integral of the input, with an inversion and scale multiplier of $1/RC$. The ability to integrate a given signal provides the analog computer with the ability to solve differential equations and therefore provides the ability to electrically solve analogs of physical system operation.

The integration operation is one of summation, summing the area under a waveform or curve over a period of time. If a fixed voltage is applied as input to an integrator circuit, Eq. (14.13) shows that the output voltage grows over a period of time, providing a ramp voltage. Equation (14.13) can thus be understood to show that the output voltage ramp (for a fixed input voltage) is opposite in polarity to the input voltage and is multiplied by the factor $1/RC$. While the circuit of Fig. 14.19 can operate on many varied types of input signals, the following examples will use only a fixed input voltage, resulting in a ramp output voltage.

As an example, consider an input voltage, $V_1 = 1$ V, to the integrator circuit of Fig. 14.20a. The scale factor of $1/RC$ is

$$-\frac{1}{RC} = \frac{1}{(1\ \text{M}\Omega)(1\ \mu\text{F})} = -1$$

so that the output is a negative ramp voltage as shown in Fig. 14.20b. If the scale factor is changed by making $R = 100$ kΩ, for example, then

$$-\frac{1}{RC} = \frac{1}{(100\ \text{k}\Omega)(1\ \mu\text{F})} = -10$$

and the output is then a steeper ramp voltage, as shown in Fig. 14.20c.

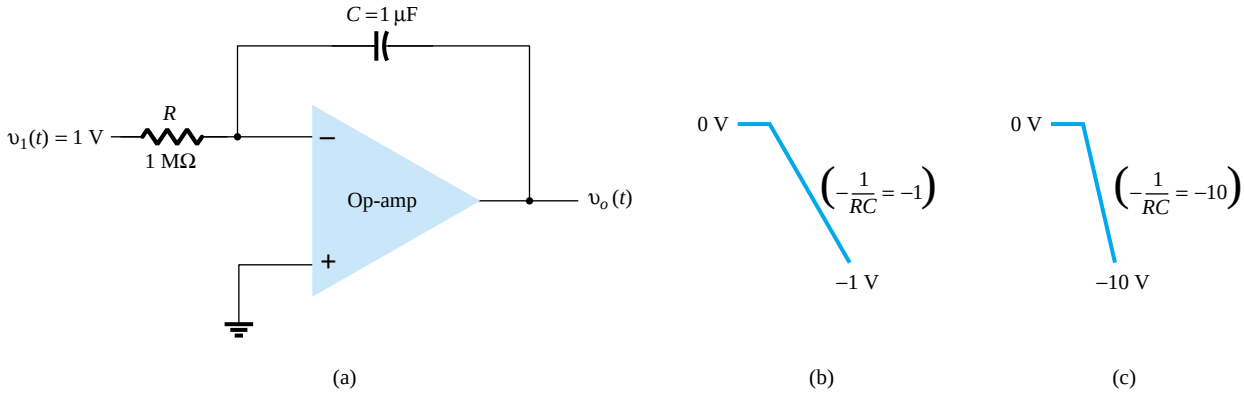


Figure 14.20 Operation of integrator with step input.

More than one input may be applied to an integrator, as shown in Fig. 14.21, with the resulting operation given by

$$v_o(t) = -\left[\frac{1}{R_1C}\int v_1(t)\,dt + \frac{1}{R_2C}\int v_2(t)\,dt + \frac{1}{R_3C}\int v_3(t)\,dt\right] \qquad (14.14)$$

An example of a summing integrator as used in an analog computer is given in Fig. 14.21. The actual circuit is shown with input resistors and feedback capacitor, whereas the analog-computer representation indicates only the scale factor for each input.

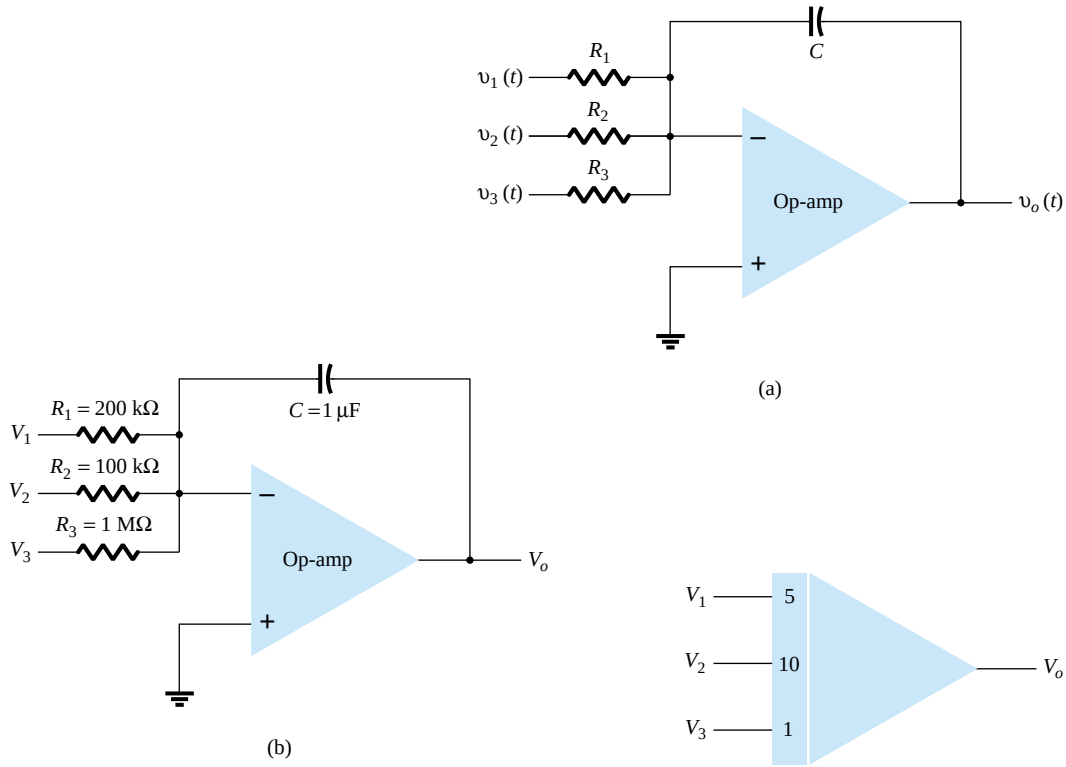


Figure 14.21 (a) Summing-integrator circuit; (b) component values; (c) analog-computer, integrator-circuit representation.

Differentiator

A differentiator circuit is shown in Fig. 14.22. While not as useful as the circuit forms covered above, the differentiator does provide a useful operation, the resulting relation for the circuit being

$$v_o(t) = -RC\frac{dv_1(t)}{dt} \tag{14.15}$$

where the scale factor is $-RC$.

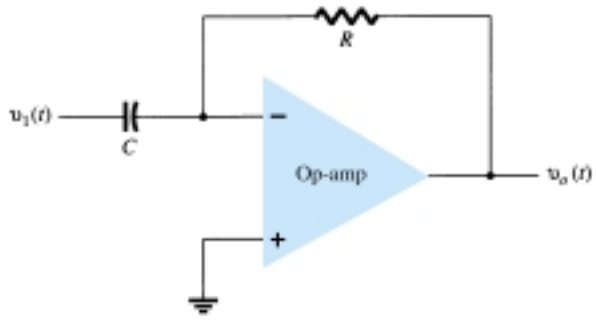


Figure 14.22 Differentiator circuit.

14.5 OP-AMP SPECIFICATIONS—DC OFFSET PARAMETERS

Before going into various practical applications using op-amps, we should become familiar with some of the parameters used to define the operation of the unit. These specifications include both dc and transient or frequency operating features, as covered next.

Offset Currents and Voltages

While the op-amp output should be 0 V when the input is 0 V, in actual operation there is some offset voltage at the output. For example, if one connected 0 V to both op-amp inputs and then measured 26 mV(dc) at the output, this would represent 26 mV of unwanted voltage generated by the circuit and not by the input signal. Since the user may connect the amplifier circuit for various gain and polarity operations, however, the manufacturer specifies an input offset voltage for the op-amp. The output offset voltage is then determined by the input offset voltage and the gain of the amplifier, as connected by the user.

The output offset voltage can be shown to be affected by two separate circuit conditions. These are: (1) an input offset voltage, V_{IO}, and (2) an offset current due to the difference in currents resulting at the plus (+) and minus (−) inputs.

INPUT OFFSET VOLTAGE, V_{IO}

The manufacturer's specification sheet provides a value of V_{IO} for the op-amp. To determine the effect of this input voltage on the output, consider the connection shown in Fig. 14.23. Using $V_o = AV_i$, we can write

$$V_o = AV_i = A\left(V_{IO} - V_o \frac{R_1}{R_1 + R_f}\right)$$

Solving for V_o, we get

$$V_o = V_{IO} \frac{A}{1 + A[R_1/(R_1 + R_f)]} \approx V_{IO} \frac{A}{A[R_1/(R_1 + R_f)]}$$

from which we can write

$$V_o(\text{offset}) = V_{IO} \frac{R_1 + R_f}{R_1} \tag{14.16}$$

Equation (14.16) shows how the output offset voltage results from a specified input offset voltage for a typical amplifier connection of the op-amp.

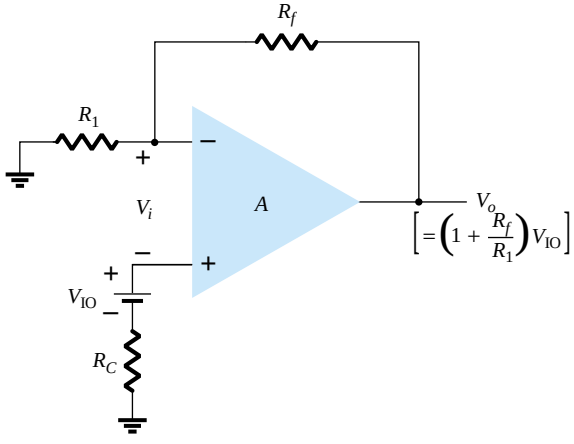


Figure 14.23 Operation showing effect of input offset voltage, V_{IO}.

EXAMPLE 14.6

Calculate the output offset voltage of the circuit in Fig. 14.24. The op-amp spec lists $V_{IO} = 1.2$ mV.

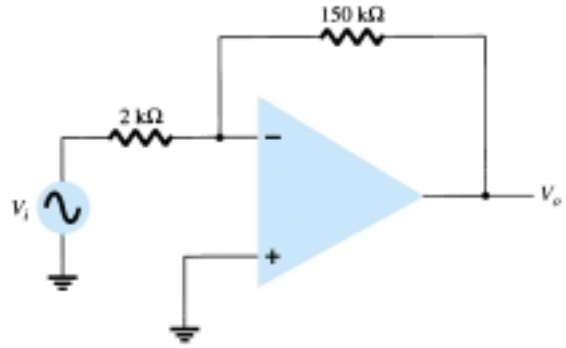


Figure 14.24 Op-amp connection for Examples 14.6 and 14.7.

Solution

$$\text{Eq. (14.16):} \quad V_o(\text{offset}) = V_{IO}\frac{R_1 + R_f}{R_1} = (1.2\text{ mV})\left(\frac{2\text{ k}\Omega + 150\text{ k}\Omega}{2\text{ k}\Omega}\right) = \mathbf{91.2\text{ mV}}$$

OUTPUT OFFSET VOLTAGE DUE TO INPUT OFFSET CURRENT, I_{IO}

An output offset voltage will also result due to any difference in dc bias currents at both inputs. Since the two input transistors are never exactly matched, each will operate at a slightly different current. For a typical op-amp connection, such as that shown in Fig. 14.25, an output offset voltage can be determined as follows. Replacing the bias currents through the input resistors by the voltage drop that each develops, as shown in Fig. 14.26, we can determine the expression for the resulting output voltage. Using superposition, the output voltage due to input bias current I_{IB}^+, denoted by V_o^+, is

$$V_o^+ = I_{IB}^+ R_C\left(1 + \frac{R_f}{R_1}\right)$$

while the output voltage due to only I_{IB}^-, denoted by V_o^-, is

$$V_o^- = I_{IB}^- R_1\left(-\frac{R_f}{R_1}\right)$$

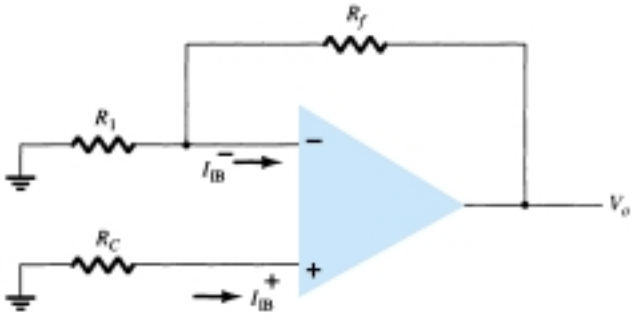


Figure 14.25 Op-amp connection showing input bias currents.

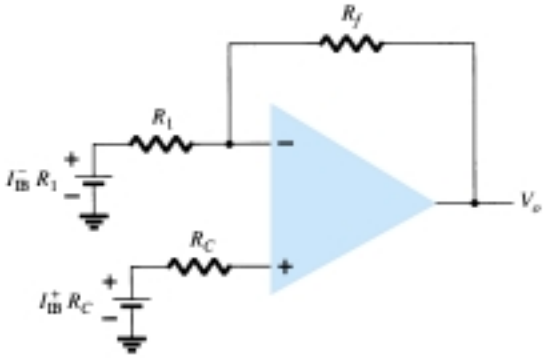


Figure 14.26 Redrawn circuit of Fig. 14.25.

for a total output offset voltage of

$$V_o(\text{offset due to } I_{\text{IB}}^{+} \text{ and } I_{\text{IB}}^{-}) = I_{\text{IB}}^{+} R_C\left(1 + \frac{R_f}{R_1}\right) - I_{\text{IB}}^{-} R_1 \frac{R_f}{R_1} \qquad (14.17)$$

Since the main consideration is the difference between the input bias currents rather than each value, we define the offset current I_{IO} by

$$I_{\text{IO}} = I_{\text{IB}}^{+} - I_{\text{IB}}^{-}$$

Since the compensating resistance R_C is usually approximately equal to the value of R_1, using $R_C = R_1$ in Eq. (14.17) we can write

$$V_o(\text{offset}) = I_{\text{IB}}^{+}(R_1 + R_f) - I_{\text{IB}}^{-} R_f$$

$$= I_{\text{IB}}^{+} R_f - I_{\text{IB}}^{-} R_f = R_f (I_{\text{IB}}^{+} - I_{\text{IB}}^{-})$$

resulting in

$$\boxed{V_o(\text{offset due to } I_{\text{IO}}) = I_{\text{IO}} R_f} \qquad (14.18)$$

EXAMPLE 14.7

Calculate the offset voltage for the circuit of Fig. 14.24 for op-amp specification listing $I_{\text{IO}} = 100$ nA.

Solution

Eq. (14.18): $V_o = I_{\text{IO}} R_f = (100 \text{ nA})(150 \text{ k}\Omega) = \mathbf{15\ mV}$

TOTAL OFFSET DUE TO V_{IO} AND I_{IO}

Since the op-amp output may have an output offset voltage due to both factors covered above, the total output offset voltage can be expressed as

$$|V_o(\text{offset})| = |V_o(\text{offset due to } V_{\text{IO}})| + |V_o(\text{offset due to } I_{\text{IO}})| \qquad (14.19)$$

The absolute magnitude is used to accommodate the fact that the offset polarity may be either positive or negative.

EXAMPLE 14.8

Calculate the total offset voltage for the circuit of Fig. 14.27 for an op-amp with specified values of input offset voltage, $V_{\text{IO}} = 4$ mV and input offset current $I_{\text{IO}} = 150$ nA.

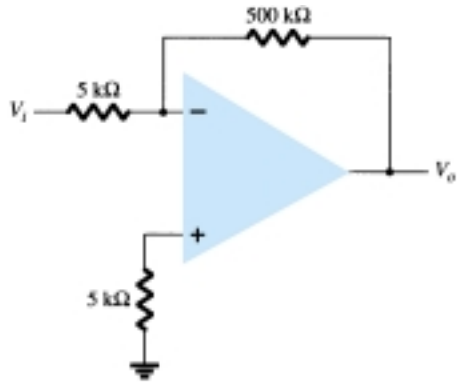
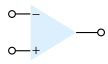


Figure 14.27 Op-amp circuit for Example 14.8.



Solution

The offset due to V_{IO} is

$$\text{Eq. (14.16):} \quad V_o(\text{offset due to } V_{IO}) = V_{IO}\frac{R_1 + R_f}{R_1} = (4\text{ mV})\left(\frac{5\text{ k}\Omega + 500\text{ k}\Omega}{5\text{ k}\Omega}\right)$$

$$= 404\text{ mV}$$

$$\text{Eq. (14.18):} \quad V_o(\text{offset due to } I_{IO}) = I_{IO}R_f = (150\text{ nA})(500\text{ k}\Omega) = 75\text{ mV}$$

resulting in a total offset

$$\text{Eq. (14.19):} \quad V_o(\text{total offset}) = V_o(\text{offset due to } V_{IO}) + V_o(\text{offset due to } I_{IO})$$

$$= 404\text{ mV} + 75\text{ mV} = \mathbf{479\text{ mV}}$$

INPUT BIAS CURRENT, I_{IB}

A parameter related to I_{IO} and the separate input bias currents I_{IB}^+ and I_{IB}^- is the average bias current defined as

$$I_{IB} = \frac{I_{IB}^+ + I_{IB}^-}{2} \tag{14.20}$$

One could determine the separate input bias currents using the specified values I_{IO} and I_{IB}. It can be shown that for $I_{IB}^+ > I_{IB}^-$

$$I_{IB}^+ = I_{IB} + \frac{I_{IO}}{2} \tag{14.21}$$

$$I_{IB}^- = I_{IB} - \frac{I_{IO}}{2} \tag{14.21}$$

EXAMPLE 14.9

Calculate the input bias currents at each input of an op-amp having specified values of $I_{IO} = 5$ nA and $I_{IB} = 30$ nA.

Solution

Using Eq. (14.21):

$$I_{IB}^+ = I_{IB} + \frac{I_{IO}}{2} = 30\text{ nA} + \frac{5\text{ nA}}{2} = \mathbf{32.5\text{ nA}}$$

$$I_{IB}^- = I_{IB} - \frac{I_{IO}}{2} = 30\text{ nA} - \frac{5\text{ nA}}{2} = \mathbf{27.5\text{ nA}}$$

14.6 OP-AMP SPECIFICATIONS—FREQUENCY PARAMETERS

An op-amp is designed to be a high-gain, wide-bandwidth amplifier. This operation tends to be unstable (oscillate) due to positive feedback (see Chapter 18). To ensure stable operation, op-amps are built with internal compensation circuitry, which also causes the very high open-loop gain to diminish with increasing frequency. This gain reduction is referred to as *roll-off*. In most op-amps, roll-off occurs at a rate of 20 dB

per decade (−20 dB/decade) or 6 dB per octave (−6 dB/octave). (Refer to Chapter 11 for introductory coverage of dB and frequency response.)

Note that while op-amp specifications list an open-loop voltage gain (A_{VD}), the user typically connects the op-amp using feedback resistors to reduce the circuit voltage gain to a much smaller value (closed-loop voltage gain, A_{CL}). A number of circuit improvements result from this gain reduction. First, the amplifier voltage gain is a more stable, precise value set by the external resistors; second, the input impedance of the circuit is increased over that of the op-amp alone; third, the circuit output impedance is reduced from that of the op-amp alone; and finally, the frequency response of the circuit is increased over that of the op-amp alone.

Gain–Bandwidth

Because of the internal compensation circuitry included in an op-amp, the voltage gain drops off as frequency increases. Op-amp specifications provide a description of the gain versus bandwidth. Figure 14.28 provides a plot of gain versus frequency for a typical op-amp. At low frequency down to dc operation the gain is that value listed by the manufacturer's specification A_{VD} (voltage differential gain) and is typically a very large value. As the frequency of the input signal increases the open-loop gain drops off until it finally reaches the value of 1 (unity). The frequency at this gain value is specified by the manufacturer as the unity-gain bandwidth, B_1. While this value is a frequency (see Fig. 14.28) at which the gain becomes 1, it can be considered a bandwidth, since the frequency band from 0 Hz to the unity-gain frequency is also a bandwidth. One could therefore refer to the point at which the gain reduces to 1 as the unity-gain frequency (f_1) or unity-gain bandwidth (B_1).

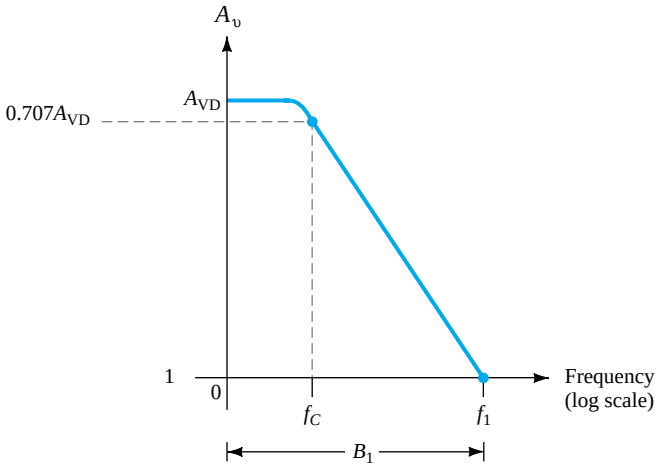
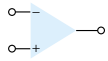


Figure 14.28 Gain versus frequency plot.

Another frequency of interest is that shown in Fig. 14.28, at which the gain drops by 3 dB (or to 0.707 the dc gain, A_{VD}), this being the cutoff frequency of the op-amp, f_C. In fact, the unity-gain frequency and cutoff frequency are related by

$$f_1 = A_{VD} f_C \tag{14.22}$$

Equation (14.22) shows that the unity-gain frequency may also be called the gain–bandwidth product of the op-amp.



EXAMPLE 14.10

Determine the cutoff frequency of an op-amp having specified values $B_1 = 1$ MHz and $A_{VD} = 200$ V/mV.

Solution

Since $f_1 = B_1 = 1$ MHz, we can use Eq. (14.22) to calculate

$$f_C = \frac{f_1}{A_{VD}} = \frac{1 \text{ MHz}}{200 \text{ V/mV}} = \frac{1 \times 10^6}{200 \times 10^3} = \mathbf{5 \text{ Hz}}$$

Slew Rate, SR

Another parameter reflecting the op-amp's ability to handling varying signals is slew rate, defined as

slew rate = maximum rate at which amplifier output can change in volts per microsecond (V/μs)

$$\boxed{\text{SR} = \frac{\Delta V_o}{\Delta t} \quad \text{V}/\mu\text{s}} \text{ with } t \text{ in } \mu\text{s} \tag{14.23}$$

The slew rate provides a parameter specifying the maximum rate of change of the output voltage when driven by a large step-input signal.* If one tried to drive the output at a rate of voltage change greater than the slew rate, the output would not be able to change fast enough and would not vary over the full range expected, resulting in signal clipping or distortion. In any case, the output would not be an amplified duplicate of the input signal if the op-amp slew rate is exceeded.

EXAMPLE 14.11

For an op-amp having a slew rate of SR = 2 V/μs, what is the maximum closed-loop voltage gain that can be used when the input signal varies by 0.5 V in 10 μs?

Solution

Since $V_o = A_{CL}V_i$, we can use

$$\frac{\Delta V_o}{\Delta t} = A_{CL}\frac{\Delta V_i}{\Delta t}$$

from which we get

$$A_{CL} = \frac{\Delta V_o/\Delta t}{\Delta V_i/\Delta t} = \frac{\text{SR}}{\Delta V_i/\Delta t} = \frac{2 \text{ V}/\mu\text{s}}{0.5 \text{ V}/10 \ \mu\text{s}} = \mathbf{40}$$

Any closed-loop voltage gain of magnitude greater than 40 would drive the output at a rate greater than the slew rate allows, so the maximum closed-loop gain is 40.

*The closed-loop gain is that obtained with the output connected back to the input in some way.

Maximum Signal Frequency

The maximum frequency that an op-amp may operate at depends on both the bandwidth (BW) and slew rate (SR) parameters of the op-amp. For a sinusoidal signal of general form

$$v_o = K \sin(2\pi f t)$$

the maximum voltage rate of change can be shown to be

$$\text{signal maximum rate of change} = 2\pi f K \qquad \text{V/s}$$

To prevent distortion at the output, the rate of change must also be less than the slew rate, that is,

$$2\pi f K \leq \text{SR}$$

$$\omega K \leq \text{SR}$$

so that

$$\boxed{\begin{aligned} f &\leq \frac{\text{SR}}{2\pi K} \quad \text{Hz} \\ \omega &\leq \frac{\text{SR}}{K} \quad \text{rad/s} \end{aligned}} \tag{14.24}$$

Additionally, the maximum frequency, f, in Eq. (14.24), is also limited by the unity-gain bandwidth.

EXAMPLE 14.12

For the signal and circuit of Fig. 14.29, determine the maximum frequency that may be used. Op-amp slew rate is SR = 0.5 V/μs.

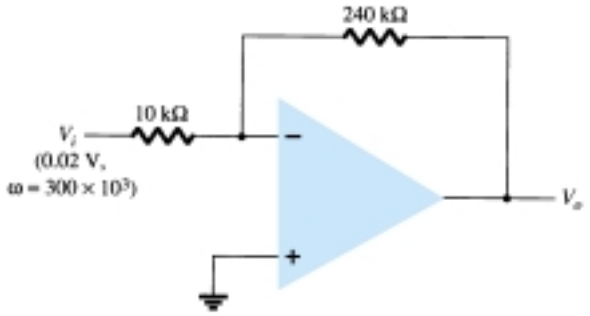


Figure 14.29 Op-amp circuit for Example 14.12.

Solution

For a gain of magnitude

$$A_{\text{CL}} = \left|\frac{R_f}{R_1}\right| = \frac{240\ \text{k}\Omega}{10\ \text{k}\Omega} = 24$$

the output voltage provides

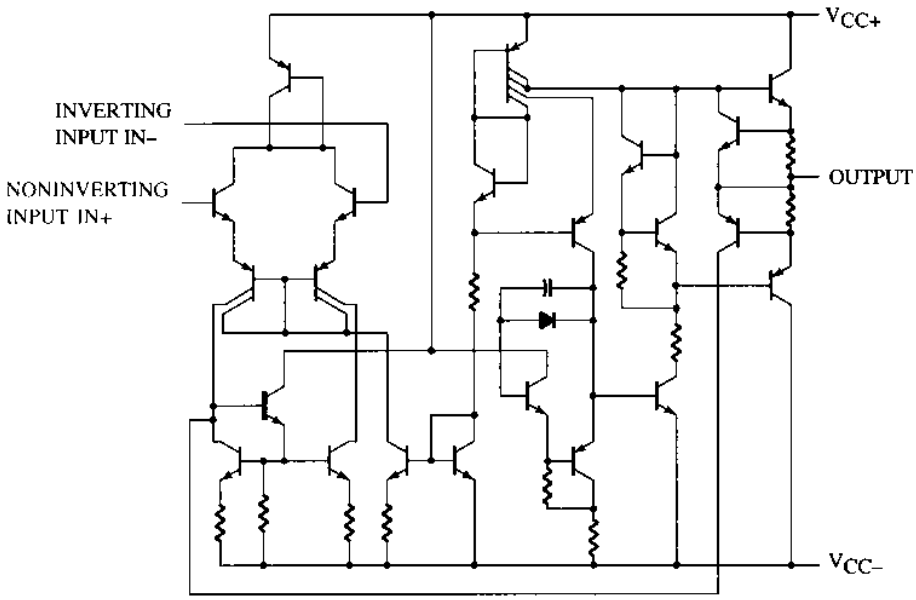
$$K = A_{\text{CL}} V_i = 24(0.02\ \text{V}) = 0.48\ \text{V}$$

$$\text{Eq. (14.24):} \quad \omega \leq \frac{\text{SR}}{K} = \frac{0.5\ \text{V}/\mu\text{s}}{0.48\ \text{V}} = \mathbf{1.1 \times 10^6\ rad/s}$$

Since the signal's frequency, $\omega = 300 \times 10^3$ rad/s, is less than the maximum value determined above, no output distortion will result.

14.7 OP-AMP UNIT SPECIFICATIONS

In this section, we discuss how the manufacturer's specifications are read for a typical op-amp unit. A popular bipolar op-amp IC is the 741 described by the information provided in Fig. 14.30. The op-amp is available in a number of packages, an 8-pin DIP and a 10-pin flatpack being among the more usual forms.



absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

		uA741M	uA741C	UNIT
Supply voltage V_{CC+} (see Note 1)		22	18	V
Supply voltage V_{CC-} (see Note 1)		– 22	– 18	V
Differential input voltage (see Note 2)		± 30	± 30	V
Input voltage any input (see Notes 1 and 3)		± 15	± 15	V
Voltage between either offset null terminal (N1/N2) and V_{CC-}		± 0.5	± 0.5	V
Duration of output short-circuit (see Note 4)		unlimited	unlimited	
Continuous total power dissipation at (or below) 25°C free-air temperature (see Note 5)		500	500	mW
Operating free-air temperature range		– 55 to 125	0 to 70	°C
Storage temperature range		– 65 to 150	– 65 to 150	°C
Lead temperature 1,6 mm (1/16 inch) from case for 60 seconds	FH, FK, J, JG, or U package	300	300	°C
Lead temperature 1,6 mm (1/16 inch) from case for 10 seconds	D, N, or P package		260	°C

NOTES: 1. All voltage values, unless otherwise noted, are with respect to the midpoint between V_{CC+} and V_{CC-}.
2. Differential voltages are at the noninverting input terminal with respect to the inverting input terminal.
3. The magnitude of the input voltage must never exceed the magnitude of the supply voltage or 15 volts, whichever is less.
4. The output may be shorted to ground or either power supply. For the uA741M only, the unlimited duration of the short-circuit applies at (or below) 125°C case temperature or 75°C free-air temperature.
5. For operation above 25°C free-air temperature, refer to Dissipation Derating Curves, Section 2. In the J and JG packages, uA741M chips are alloy mounted; uA741C chips are glass mounted.

Figure 14.30 741 op-amp specifications.

electrical characteristics at specified free-air temperature, V_{CC+} = 15 V, V_{CC-} = −15 V

PARAMETER		TEST CONDITIONS†		μA741M			μA741C			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
V_{IO}	Input offset voltage	V_O = 0	25°C		1	5		1	6	mV
			Full range			6			7.5	
$\Delta V_{IO(adj)}$	Offset voltage adjust range	V_O = 0	25°C		±15			±15		mV
I_{IO}	Input offset current	V_O = 0	25°C		20	200		20	200	nA
			Full range			500			300	
I_{IB}	Input bias current	V_O = 0	25°C		80	500		80	500	nA
			Full range			1500			800	
V_{ICR}	Common-mode input voltage range		25°C	±12	±13		±12	±13		V
			Full range	±12			±12			
V_{OM}	Maximum peak output voltage swing	R_L = 10 kΩ	25°C	±12	±14		±12	±14		V
		R_L ≥ 10 kΩ	Full range	±12			±12			
		R_L = 2 kΩ	25°C	±10	±13		±10	±13		
		R_L ≥ 2 kΩ	Full range	±10			±10			
A_{VD}	Large-signal differential voltage amplification	R_L ≥ 2 kΩ	25°C	50	200		20	200		V/mV
		V_O = ±10 V	Full range	25			15			
r_i	Input resistance		25°C	0.3	2		0.3	2		MΩ
r_o	Output resistance	VO = 0 See note 6	25°C		75			75		Ω
C_i	Input capacitance		25°C		1.4			1.4		pF
CMRR	Common-mode rejection ratio	V_{IC} = V_{ICR} min	25°C	70	90		70	90		dB
			Full range	70			70			
k_{SVS}	Supply voltage sensitivity $\Delta V_{IO}/\Delta V_{CC}$)	V_{CC} = ±9 V to ±15 V	25°C		30	150		30	150	μV/V
			Full range			150			150	
I_{OS}	Short-circuit output current		25°C		±25	±40		±25	±40	mA
I_{CC}	Supply current	No load, V_O = 0	25°C		1.7	2.8		1.7	2.8	mA
			Full range			3.3			3.3	
P_D	Total power dissipation	No load, V_O = 0	25°C		50	85		50	85	mW
			Full range			100			100	

operating characteristics, V_{CC+} = 15 V, V_{CC-} = −15 V, T_A = 25°C

PARAMETER		TEST CONDITIONS	μA741M			μA741C			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
t_r	Rise time	V_I = 20 mV, R_L = 2 kΩ, C_L = 100 pF, See Figure 1		0.3			0.3		μs
	Overshoot factor			5%			5%		
SR	Slew rate at unity gain	V_I = 10 V, R_L = 2 kΩ, C_L = 100 pF, See Figure 1		0.5			0.5		V/μs

Figure 14.30 Continued.

Absolute Maximum Ratings

The absolute maximum ratings provide information on what largest voltage supplies may be used, how large the input signal swing may be, and at how much power the device is capable of operating. Depending on the particular version of 741 used, the largest supply voltage is a dual supply of ±18 V or ±22 V. In addition, the IC can internally dissipate from 310 to 570 mW, depending on the IC package used. Table 14.1 summarizes some typical values to use in examples and problems.

TABLE 14.1 Absolute Maximum Ratings

Supply voltage	±22 V
Internal power dissipation	500 mW
Differential input voltage	±30 V
Input voltage	±15 V

EXAMPLE 14.13

Determine the current draw from a dual power supply of ±12 V if the IC dissipates 500 mW.

Solution

If we assume that each supply provides half the total power to the IC, then

$$P = VI$$

$$250 \text{ mW} = 12 \text{ V}(I)$$

so that each supply must provide a current of

$$I = \frac{250 \text{ mW}}{12 \text{ V}} = \mathbf{20.83 \text{ mA}}$$

Electrical Characteristics

Electrical characteristics include many of the parameters covered earlier in this chapter. The manufacturer provides some combination of typical, minimum, or maximum values for various parameters as deemed most useful to the user. A summary is provided in Table 14.2.

TABLE 14.2 μA741 Electrical Characteristics: $V_{CC} = \pm 15$ V, $T_A = 25°C$

Characteristic	*MIN*	*TYP*	*MAX*	*Unit*
V_{IO} Input offset voltage		1	6	mV
I_{IO} Input offset current		20	200	nA
I_{IB} Input bias current		80	500	nA
V_{ICR} Common-mode input voltage range	±12	±13		V
V_{OM} Maximum peak output voltage swing	±12	±14		V
A_{VD} Large-signal differential voltage amplification	20	200		V/mV
r_i Input resistance	0.3	2		MΩ
r_o Output resistance		75		Ω
C_i Input capacitance		1.4		pF
CMRR Common-mode rejection ratio	70	90		dB
I_{CC} Supply current		1.7	2.8	mA
P_D Total power dissipation		50	85	mW

V_{IO} Input offset voltage: The input offset voltage is seen to be typically 1 mV, but can go as high as 6 mV. The output offset voltage is then computed based on the circuit used. If the worst condition possible is of interest, the maximum value should be used. Typical values are those more commonly expected when using the op-amp.

I_{IO} Input offset current: The input offset current is listed to be typically 20 nA, while the largest value expected is 200 nA.

I_{IB} Input bias current: The input bias current is typically 80 nA and may be as large as 500 nA.

V_{ICR} Common-mode input voltage range: This parameter lists the range that the input voltage may vary over (using a supply of ±15 V), about ±12 to ±13 V. Inputs larger in amplitude than this value will probably result in output distortion and should be avoided.

V_{OM} Maximum peak output voltage swing: This parameter lists the largest value the output may vary (using a ±15-V supply). Depending on the circuit closed-

loop gain, the input signal should be limited to keep the output from varying by an amount no larger than ±12 V, in the worst case, or by ±14 V, typically.

A_{VD} **Large-signal differential voltage amplification:** This is the open-loop voltage gain of the op-amp. While a minimum value of 20 V/mV or 20,000 V/V is listed, the manufacturer also lists a typical value of 200 V/mV or 200,000 V/V.

r_i **Input resistance:** The input resistance of the op-amp when measured under open-loop is typically 2 MΩ but could be as little as 0.3 MΩ or 300 kΩ. In a closed-loop circuit, this input impedance can be much larger, as discussed previously.

r_o **Output resistance:** The op-amp output resistance is listed as typically 75 Ω. No minimum or maximum value is given by the manufacturer for this op-amp. Again, in a closed-loop circuit, the output impedance can be lower, depending on the circuit gain.

C_i **Input capacitance:** For high-frequency considerations, it is helpful to know that the input to the op-amp has typically 1.4 pF of capacitance, a generally small value compared even to stray wiring.

CMRR Common-mode rejection ratio: The op-amp parameter is seen to be typically 90 dB but could go as low as 70 dB. Since 90 dB is equivalent to 31622.78, the op-amp amplifies noise (common inputs) by over 30,000 times less than difference inputs.

I_{CC} **Supply current:** The op-amp draws a total of 2.8 mA, typically from the dual voltage supply, but the current drawn could be as little as 1.7 mA. This parameter helps the user determine the size of the voltage supply to use. It also can be used to calculate the power dissipated by the IC ($P_D = 2V_{CC}I_{CC}$).

P_D **Total power dissipation:** The total power dissipated by the op-amp is typically 50 mW but could go as high as 85 mW. Referring to the previous parameter, the op-amp will dissipate about 50 mW when drawing about 1.7 mA using a dual 15-V supply. At smaller supply voltages, the current drawn will be less and the total power dissipated will also be less.

EXAMPLE 14.14

Using the specifications listed in Table 14.2, calculate the typical output offset voltage for the circuit connection of Fig. 14.31.

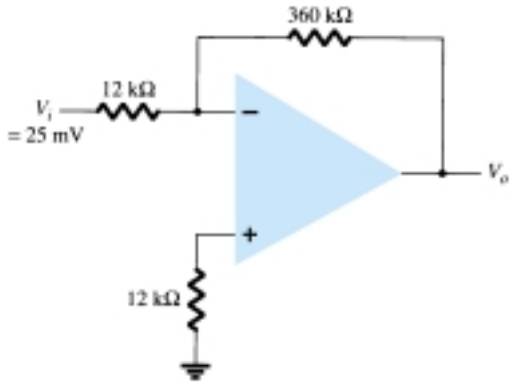


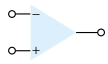
Figure 14.31 Op-amp circuit for Examples 14.14, 14.15, and 14.17.

Solution

The output offset due to V_{IO} is calculated to be

$$\text{Eq. (14.16):} \quad V_o(\text{offset}) = V_{IO}\frac{R_1 + R_f}{R_1} = (1\text{ mV})\left(\frac{12\text{ k}\Omega + 360\text{ k}\Omega}{12\text{ k}\Omega}\right) = 31\text{ mV}$$

The output voltage due to I_{IO} is calculated to be



Eq. (14.18): $V_o(\text{offset}) = I_{IO}R_f = 20\text{ nA}(360\text{ k}\Omega) = 7.2\text{ mV}$

Assuming that these two offsets are the same polarity at the output, the total output offset voltage is then

$$V_o(\text{offset}) = 31\text{ mV} + 7.2\text{ mV} = \mathbf{38.2\text{ mV}}$$

EXAMPLE 14.15

For the typical characteristics of the 741 op-amp ($r_o = 75\ \Omega$, $A = 200\text{ k}\Omega$), calculate the following values for the circuit of Fig. 14.31.

(a) A_{CL}.
(b) Z_i.
(c) Z_o.

Solution

(a) Eq. (14.8): $$\frac{V_o}{V_i} = -\frac{R_f}{R_1} = -\frac{360\text{ k}\Omega}{12\text{ k}\Omega} = -\mathbf{30} \cong \frac{1}{\beta}$$

(b) $Z_i = R_1 = \mathbf{12\text{ k}\Omega}$

(c) $$Z_o = \frac{r_o}{(1+\beta A)} = \frac{75\ \Omega}{1 + \left(\frac{1}{30}\right)(200\text{ k}\Omega)} = \mathbf{0.011\ \Omega}$$

Operating Characteristics

Another group of values used to describe the operation of the op-amp over varying signals are provided in Table 14.3.

TABLE 14.3 Operating Characteristics: $V_{CC} = \pm 15$ V, $T_A = 25°C$

Parameter	*MIN*	*TYP*	*MAX*	*Unit*
B_1 Unity gain bandwidth		1		MHz
t_r Rise time		0.3		μs

EXAMPLE 14.16

Calculate the cutoff frequency of an op-amp having characteristics given in Tables 14.2 and 14.3.

Solution

$$\text{Eq. (14.22): } f_C = \frac{f_1}{A_{VD}} = \frac{B_1}{A_{VD}} = \frac{1\text{ MHz}}{20{,}000} = \mathbf{50\text{ Hz}}$$

EXAMPLE 14.17

Calculate the maximum frequency of the input signal for the circuit in Fig. 14.31, with an input of $V_i = 25$ mV.

Solution

For a closed-loop gain of $A_{CL} = 30$ and an input of $V_i = 25$ mV, the output gain factor is calculated to be

$$K = A_{CL}V_i = 30(25\text{ mV}) = 750\text{ mV} = 0.750\text{ V}$$

Using Eq. (14.24), the maximum signal frequency, f_{max}, is

$$f_{max} = \frac{SR}{2\pi K} = \frac{0.5\ V/\mu s}{2\pi(0.750\ V)} = \mathbf{106\ kHz}$$

Op-Amp Performance

The manufacturer provides a number of graphical descriptions to describe the performance of the op-amp. Figure 14.32 includes some typical performance curves comparing various characteristics as a function of supply voltage. The open-loop voltage gain is seen to get larger with a larger supply voltage value. While the previous tabular information provided information at a particular supply voltage, the performance curve shows how the voltage gain is affected by using a range of supply voltage values.

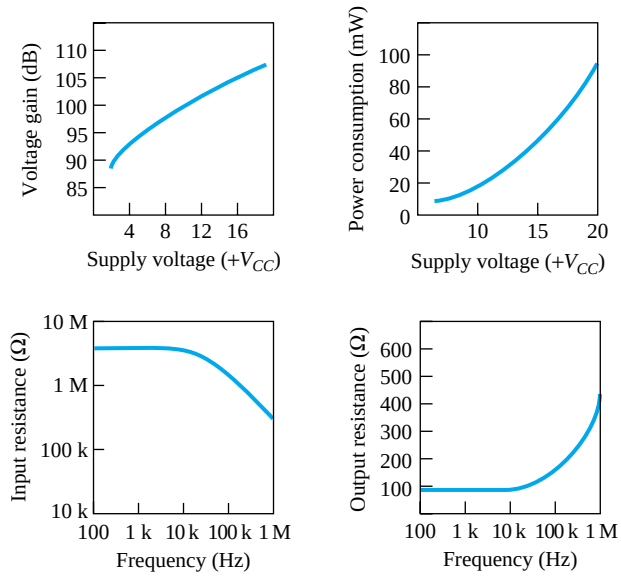


Figure 14.32 Performance curves.

EXAMPLE 14.18

Using Fig. 14.32, determine the open-loop voltage gain for a supply voltage of $V_{CC} = \pm 12$ V.

Solution

From the curve in Fig. 14.32, $A_{VD} \approx 104$ dB. This is a linear voltage gain of

$$A_{VD}\ (\text{dB}) = 20 \log_{10} A_{VD}$$

$$104\ \text{dB} = 20 \log A_{VD}$$

$$A_{VD} = \text{antilog}\ \frac{104}{20} = 158.5 \times 10^3$$

Another performance curve in Fig. 14.32 shows how power consumption varies as a function of supply voltage. As shown, the power consumption increases with larger values of supply voltage. For example, while the power dissipation is about 50 mW at $V_{CC} = \pm 15$ V, it drops to about 5 mW with $V_{CC} = \pm 5$ V. Two other curves show how the input and output resistances are affected by frequency, the input impedance dropping and the output resistance increasing at higher frequency.

14.8 PSPICE WINDOWS

The evaluation version of PSpice has only four op-amp units. These are defined by the subcircuit made up of various transistors, resistors, capacitors, and so on. These are models of four of the more common op-amp units and have their unit specifications. One can model an op-amp to provide a more ideal unit—this being helpful when describing theoretical circuit connections. Lets start by describing an op-amp model that can be used to analyze circuits.

PSpice Op-Amp Model

An op-amp can be described by a schematic circuit having an input impedance, R_i, an output impedance, R_O, and a voltage gain, A_v. Figure 14.33 shows this basic circuit, using the typical values of a 741 op-amp:

$$R_i = 2\ \text{M}\Omega, \qquad R_O = 75\Omega, \qquad A_v = 200{,}000 = 200\ \text{V/mV}$$

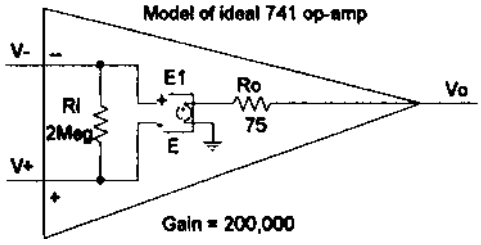


Figure 14.33 PSpice ideal op-amp model.

The values of input and output resistance are provided by resistor components with desired values. The gain of the op-amp is provided using a voltage-controlled voltage source, schematic device part labeled **E.** Figure 14.34 shows setting the E device for a gain of 200,000 (the device parameter **GAIN** is set to a value of 200,000). The schematic circuit of Fig. 14.33 thus represents a 741 op-amp with typical specs listed above.

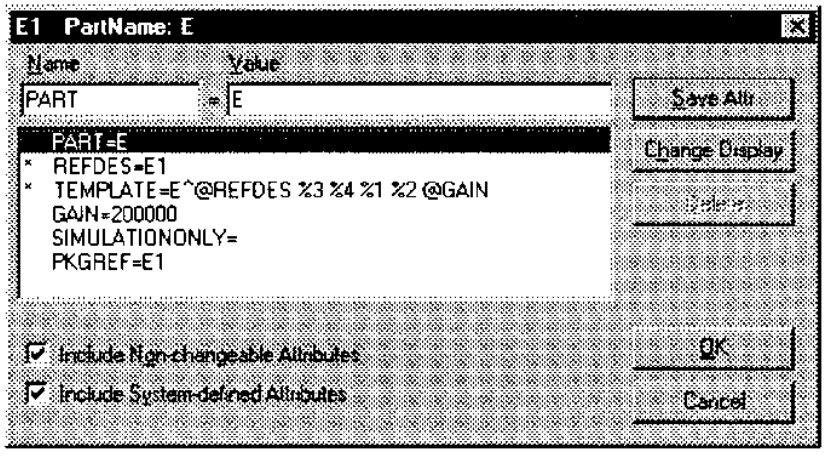


Figure 14.34 Setting gain of part E.

Program 14.1—Inverting Op-Amp

An inverting op-amp of the type described in Example 14.3 and shown in Fig. 14.15 is considered first. Using the ideal model of Fig. 14.33, an inverting op-amp circuit is drawn as in Fig. 14.35. With the dc voltage display turned on, the result after running an analysis shows that for an input of 2 V and a circuit gain of −5.

$$A_v = -R_F/R_1 = -500\ \text{k}\Omega/100\ \text{k}\Omega = -5$$

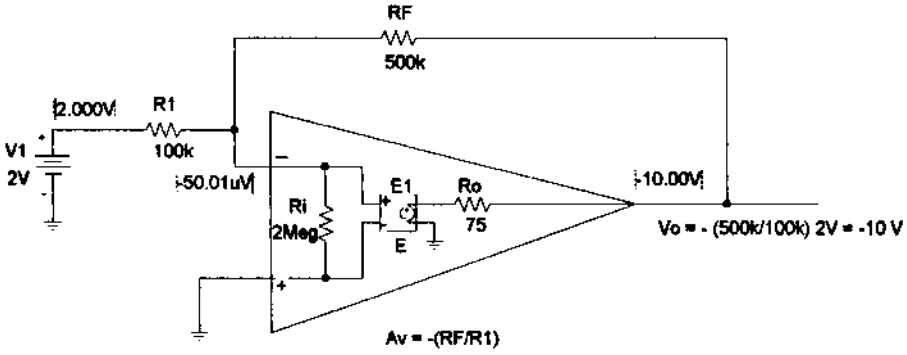


Figure 14.35 Inverting op-amp using ideal model.

The output is exactly −10 V

$$V_O = A_v V_i = -5(2\ \text{V}) = -10\ \text{V}$$

The input to the minus terminal is −50.01 μV, which is virtually ground or 0 V.

A practical inverting op-amp circuit is drawn in Fig. 14.36. Using the same resistor values as in Fig. 14.35 with a practical op-amp unit, the μA741, the resulting output is −9.96 V, near the ideal value of −10 V. This slight difference from the ideal is due to the actual gain and input impedance of the μA741 op-amp unit. Fig. 14.36 shows dc voltages because the **Enable Bias Voltage Display** was set on. Notice the minus input is 69.26 μV for this op-amp circuit—slightly different from that using the op-amp model of Fig. 14.33.

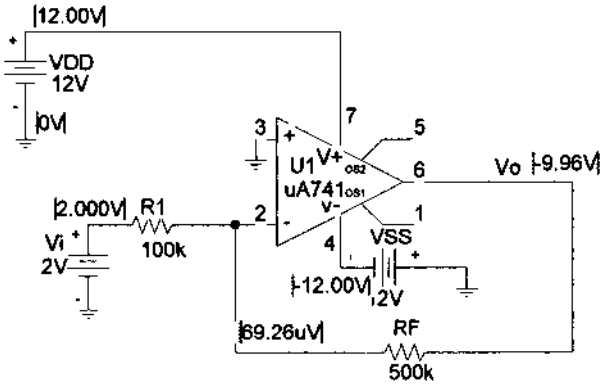


Figure 14.36 Practical inverting op-amp circuit.

An output listing from the analysis of Fig. 14.36 is shown in Fig. 14.37. Before the analysis is done, selecting **Analysis Setup, Transfer Function,** and then **Output** of **V(RF:2)** and **Input Source** of V_i will provide the small-signal characteristics in the output listing. The circuit gain is seen to be

```
Inverting op-amp circuit

****     SMALL-SIGNAL CHARACTERISTICS
  V(Vo)/V_Vi = -5.000E+00
  INPUT RESISTANCE AT V_Vi =  1.000E+05
  OUTPUT RESISTANCE AT V(Vo) =  4.950E-03
```

Figure 14.37 PSpice output for inverting op-amp (edited).

$V_O/V_i = -5$

Input resistance at $V_i = 1 \times 10^5$

Output resistance at $V_O = 4.95 \times 10^{-3}$

Program 14.2—Noninverting Op-Amp

Fig. 14.38 shows a noninverting op-amp circuit. The bias voltages are displayed on the figure. The theoretical gain of the amplifier circuit should be

$$A_v = (1 + RF/R1) = 1 + 500\ \text{k}\Omega/100\ \text{k}\Omega = 6$$

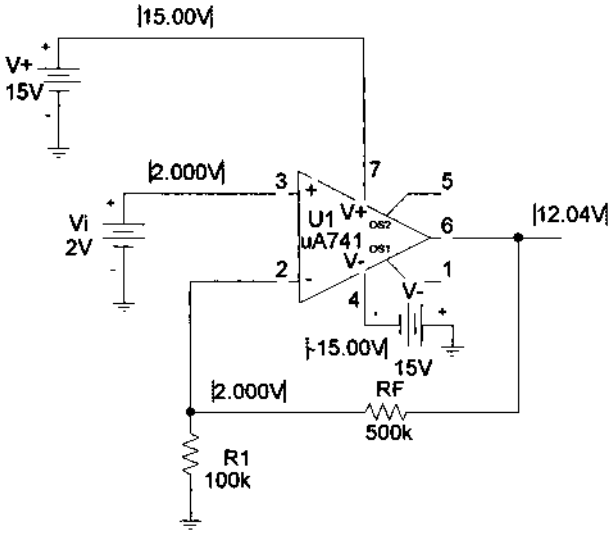


Figure 14.38 Design Center schematic for noninverting op-amp circuit.

For an input of 2 V, the resulting output will be

$$V_O = A_v V_i = 5(2\ \text{V}) = 10\ \text{V}$$

The output is noninverted from the input.

Program 14.3—Summing Op-Amp Circuit

A summing op-amp circuit such as that in Example 14.5 is shown in Fig. 14.39. Bias voltages also are displayed in Fig. 14.39, showing the resulting output at 3 V, as was

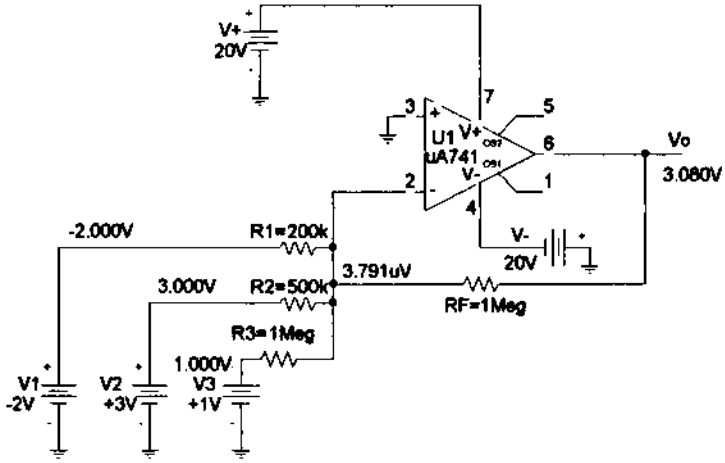


Figure 14.39 Summing amplifier for Program 14.3.

calculated in Example 14.5. Notice how well the virtual ground concept works with the minus input being only 3.791 μV.

Program 14.4—Unity-Gain Op-Amp Circuit

Figure 14.40 shows a unity-gain op-amp circuit with bias voltages displayed. For an input of +2 V, the output is exactly +2 V.

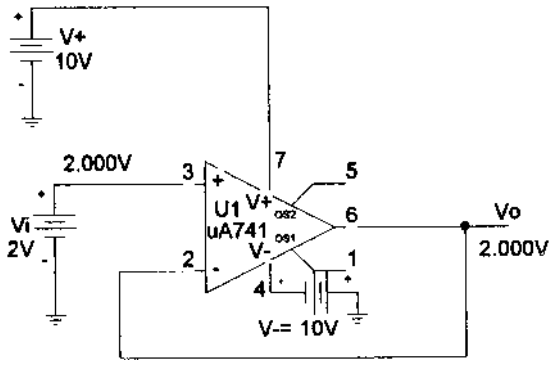


Figure 14.40 Unity-gain amplifier.

Program 14.5—Op-Amp Integrator Circuit

An op-amp integrator circuit is shown in Fig. 14.41. The input is selected as **VPULSE,** which is set to be a step input as follows:

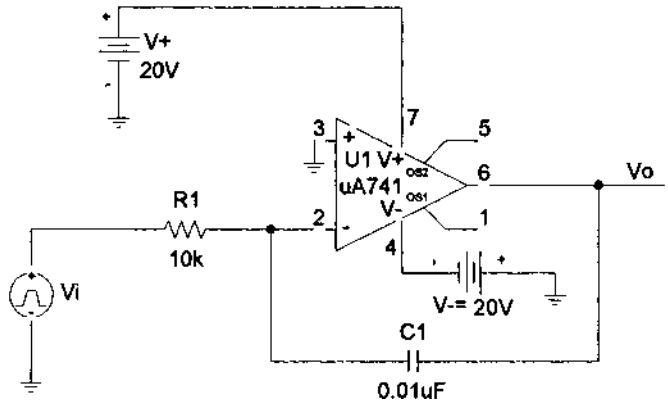


Figure 14.41 Op-amp integrator circuit.

Set **ac** = 0, **dc** = 0, **V1** = 0 V, **V2** = 2 V, **TD** = 0, **TR** = 0, **TF** = 0, **PW** = 10 ms, and **PER** = 20 ms. This provides a step from 0 to 2 V, with no time delay, rise time or fall time, having a period of 10 ms and repeating after a period of 20 ms. For this problem, the voltage rises instantly to 2 V, then stays there for a sufficiently long time for the output to drop as a ramp voltage from the maximum supply level of +20 V to the lowest level of −20 V. Theoretically, the output for the circuit of Fig. 14.41 is

$$v_O(t) = -1/RC \int v_i(t)\, dt$$

$$v_O(t) = -1/(10\ \text{k}\Omega)(0.01\ \mu\text{F}) \int 2\, dt = -10{,}000 \int 2\, dt = -20{,}000t$$

This is a negative ramp voltage dropping at a rate (slope) of −20,000 V/s. This ramp voltage will drop from +20 V to −20 V in

$$40 \text{ V}/20{,}000 = 2 \times 10^{-3} = 2 \text{ ms}$$

Fig. 14.42 shows the input step waveform and the resulting output ramp waveform obtained using **PROBE.**

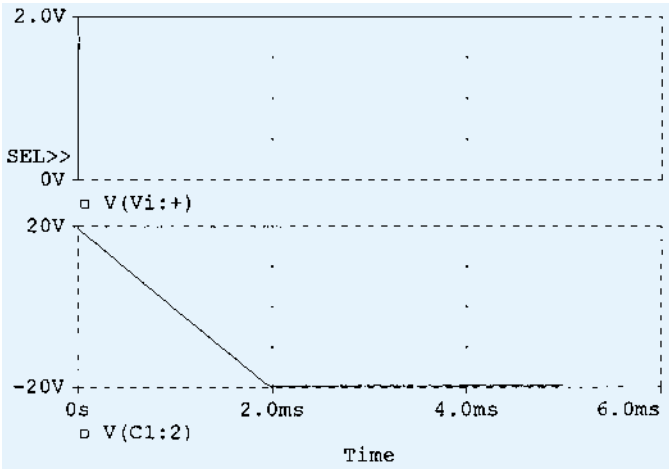


Figure 14.42 Probe waveform for integrator circuit.

Program 14.6—Multistage Op-Amp Circuit

A multistage op-amp circuit is shown in Fig. 14.43. The input to stage 1 of 200 mV provides an output of 200 mV to stages 2 and 3. Stage 2 is an inverting amplifier with

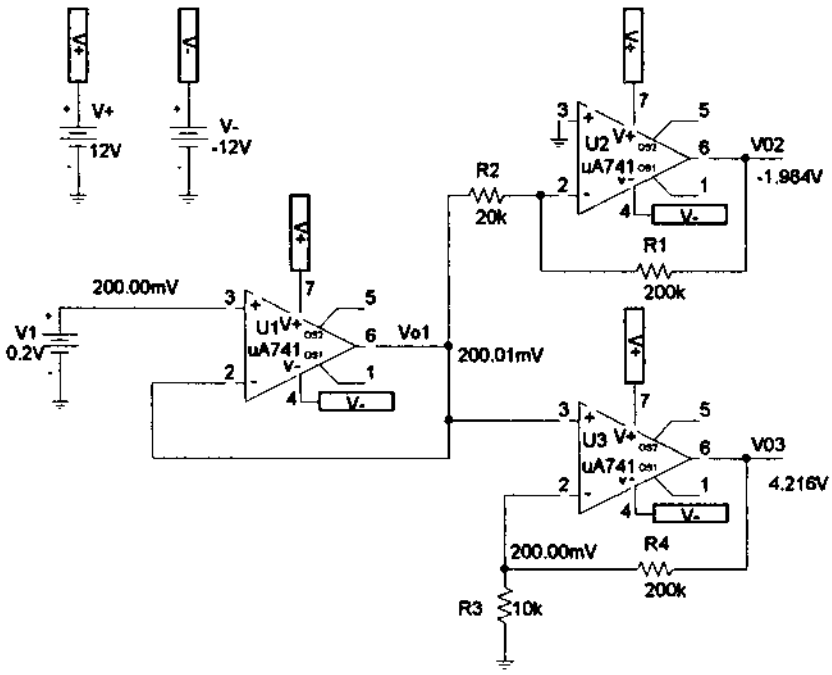


Figure 14.43 Multistage op-amp circuit.

gain $-200\ \text{k}\Omega/20\ \text{k}\Omega = -10$, with an output from stage 2 of $-10(200\ \text{mV}) = -2$ V. State 3 is a non-inverting amplifier with gain of $(1 + 200\ \text{k}\Omega/10\ \text{k}\Omega = 21)$, resulting in an output of $21(200\ \text{mV}) = 4.2$ V.

PROBLEMS

§ 14.2 Differential and Common-Mode Operation

1. Calculate the CMRR (in dB) for the circuit measurements of $V_d = 1$ mV, $V_o = 120$ mV, and $V_C = 1$ mV, $V_o = 20\ \mu$V.

2. Determine the output voltage of an op-amp for input voltages of $V_{i_1} = 200\ \mu$V and $V_{i_2} = 140\ \mu$V. The amplifier has a differential gain of $A_d = 6000$ and the value of CMRR is:
(a) 200.
(b) 10^5.

§ 14.4 Practical Op-Amp Circuits

3. What is the output voltage in the circuit of Fig. 14.44?

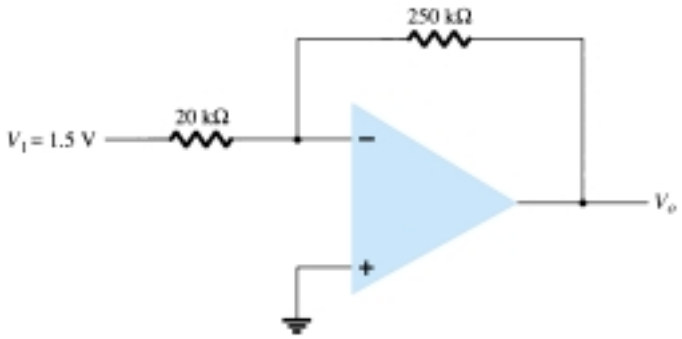


Figure 14.44 Problems 3 and 25

4. What is the range of the voltage-gain adjustment in the circuit of Fig. 14.45?

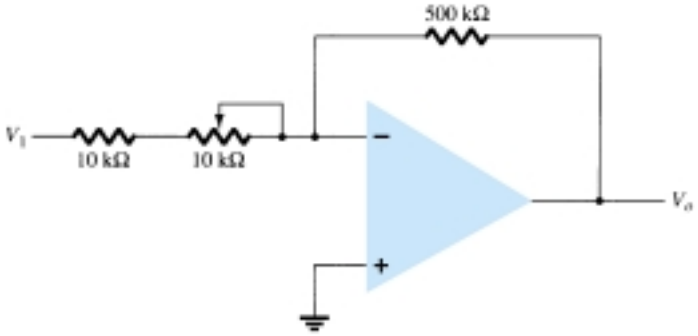


Figure 14.45 Problem 4

5. What input voltage results in an output of 2 V in the circuit of Fig. 14.46?

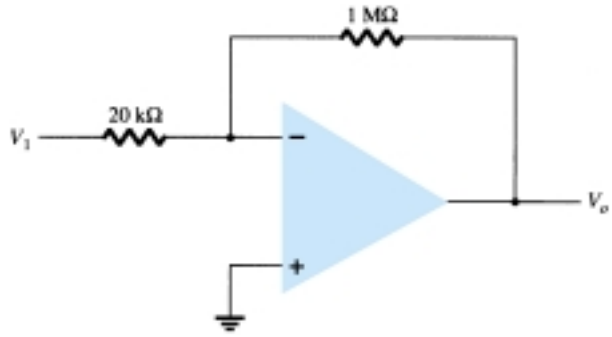


Figure 14.46 Problem 5

6. What is the range of the output voltage in the circuit of Fig. 14.47 if the input can vary from 0.1 to 0.5 V?

7. What output voltage results in the circuit of Fig. 14.48 for an input of $V_1 = -0.3$ V?

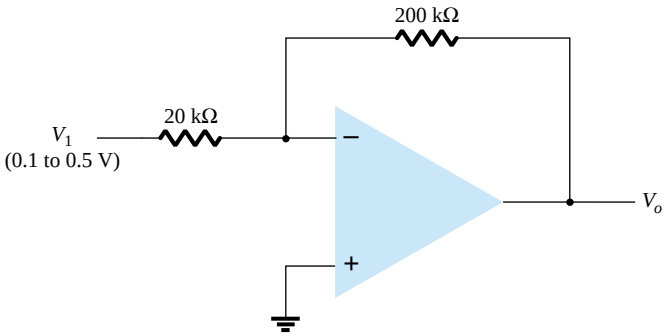


Figure 14.47 Problem 6

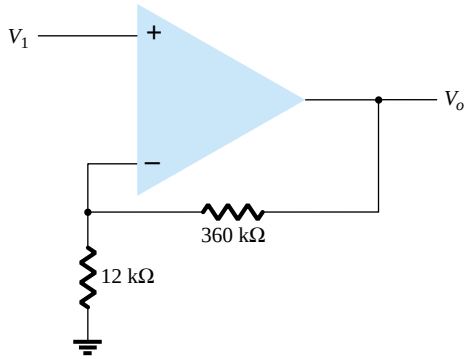


Figure 14.48 Problems 7, 8, and 26

8. What input must be applied to the input of Fig. 14.48 to result in an output of 2.4 V?

9. What range of output voltage is developed in the circuit of Fig. 14.49?

10. Calculate the output voltage developed by the circuit of Fig. 14.50 for $R_f = 330$ kΩ.

11. Calculate the output voltage of the circuit in Fig. 14.50 for $R_f = 68$ kΩ.

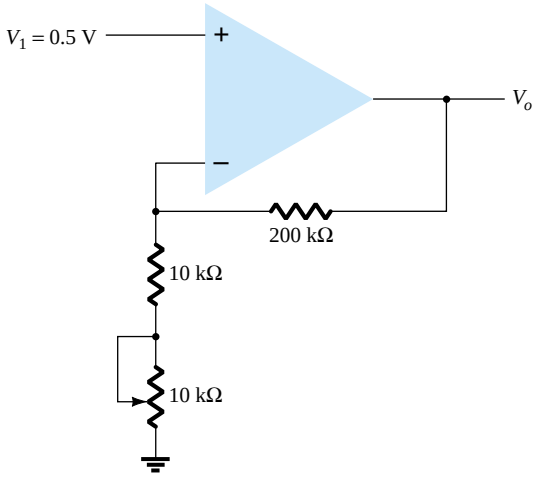


Figure 14.49 Problem 9

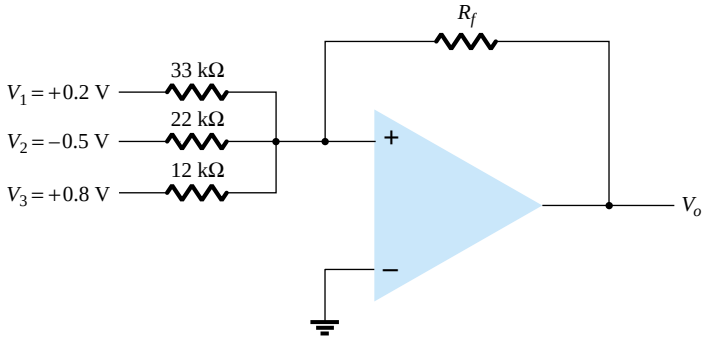


Figure 14.50 Problems 10, 11, and 27

12. Sketch the output waveform resulting in Fig. 14.51.

13. What output voltage results in the circuit of Fig. 14.52 for $V_1 = +0.5$ V?

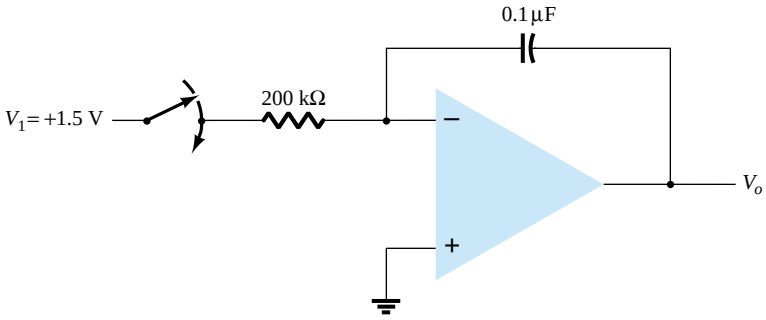


Figure 14.51 Problem 12

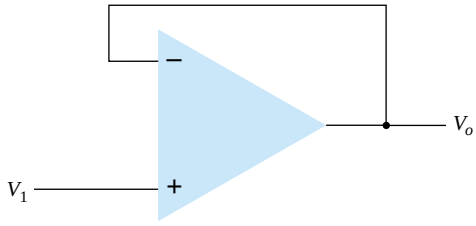


Figure 14.52 Problem 13

14. Calculate the output voltage for the circuit of Fig. 14.53.

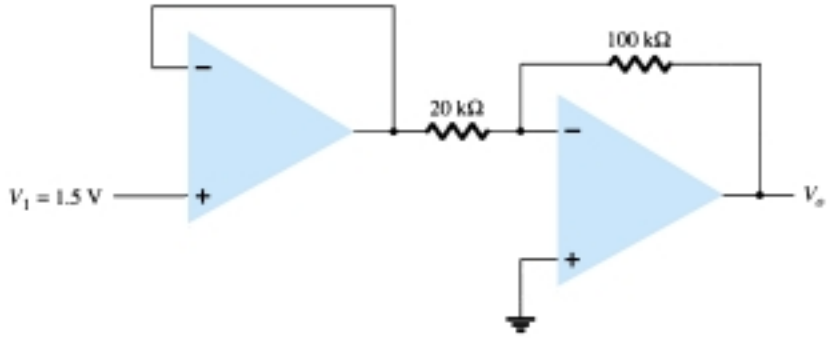


Figure 14.53 Problems 14 and 28

15. Calculate the output voltages V_2 and V_3 in the circuit of Fig. 14.54.

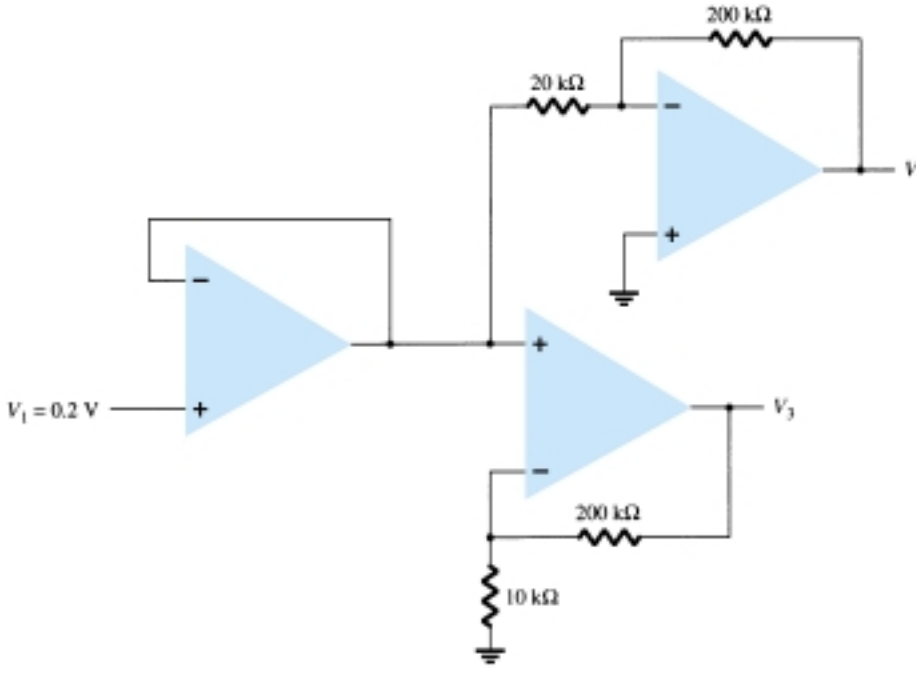


Figure 14.54 Problem 15

16. Calculate the output voltage, V_o, in the circuit of Fig. 14.55.

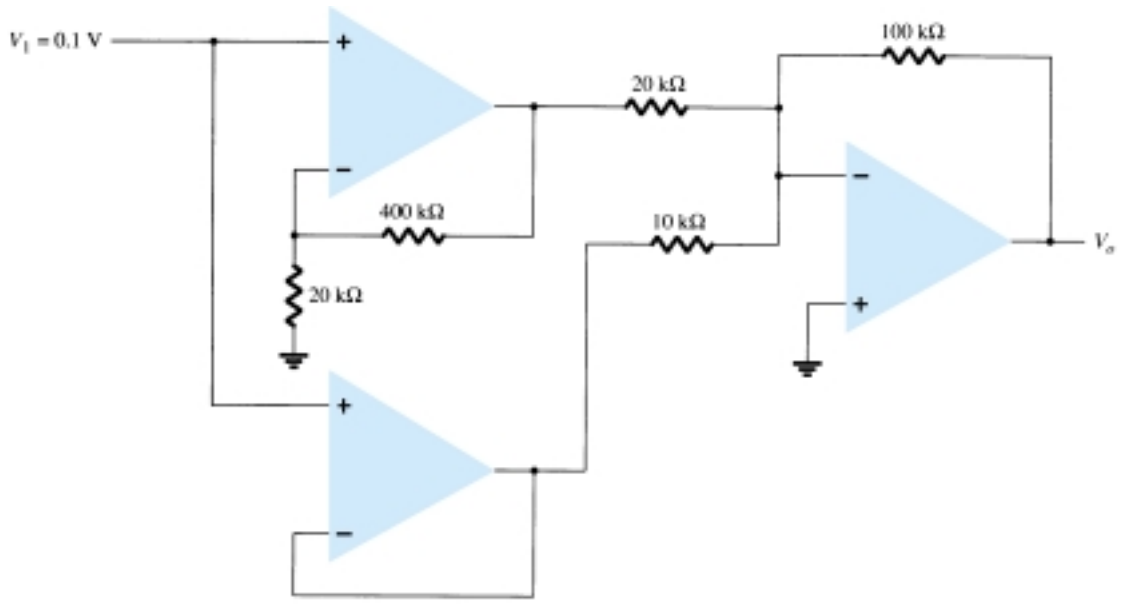


Figure 14.55 Problems 16 and 29

17. Calculate V_o in the circuit of Fig. 14.56.

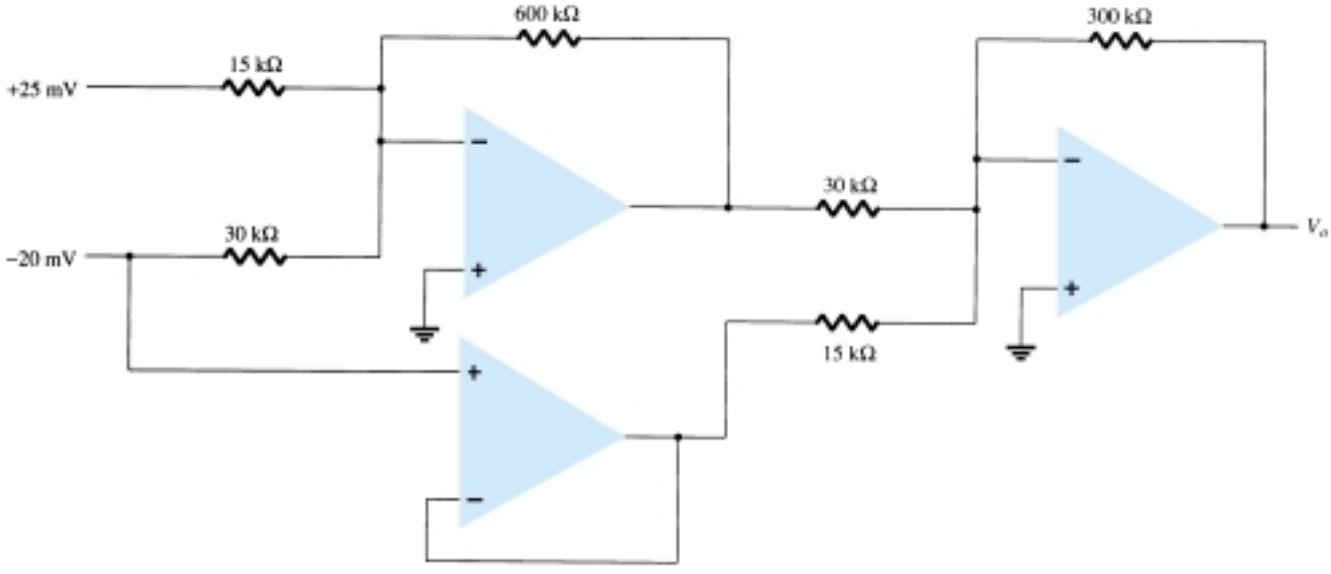


Figure 14.56 Problem 17

§ 14.5 Op-Amp Specifications—DC Offset Parameters

* **18.** Calculate the total offset voltage for the circuit of Fig. 14.57 for an op-amp with specified values of input offset voltage $V_{IO} = 6$ mV and input offset current $I_{IO} = 120$ nA.

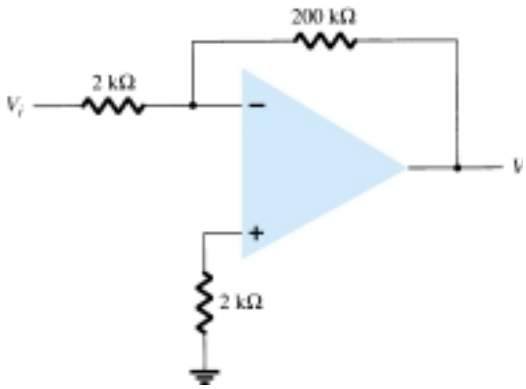


Figure 14.57 Problems 18, 22, 23, and 24

* **19.** Calculate the input bias current at each input of an op-amp having specified values of I_{IO} = 4 nA and I_{IB} = 20 nA.

§ 14.6 Op-Amp Specifications—Frequency Parameters

20. Determine the cutoff frequency of an op-amp having specified values B_1 = 800 kHz and A_{VD} = 150 V/mV.

* **21.** For an op-amp having a slew rate of SR = 2.4 V/μs, what is the maximum closed-loop voltage gain that can be used when the input signal varies by 0.3 V in 10 μs?

* **22.** For an input of V_1 = 50 mV in the circuit of Fig. 14.57, determine the maximum frequency that may be used. The op-amp slew rate SR = 0.4 V/μs.

23. Using the specifications listed in Table 14.2, calculate the typical offset voltage for the circuit connection of Fig. 14.57.

* **24.** For the typical characteristics of the 741 op-amp, calculate the following values for the circuit of Fig. 14.57.
(a) A_{CL}.
(b) Z_i.
(c) Z_o.

§ 14.8 PSpice Windows

* **25.** Use Schematic Capture to draw a circuit to determine the output voltage in the circuit of Fig. 14.44.

* **26.** Use Schematic Capture to calculate the output voltage in the circuit of Fig. 14.48 for the input of V_i = 0.5 V.

* **27.** Use Schematic Capture to calculate the output voltage in the circuit of Fig. 14.50 for R_f = 68 kΩ.

* **28.** Use Schematic Capture to calculate the output voltage in the circuit of Fig. 14.53.

* **29.** Use Schematic Capture to calculate the output voltage in the circuit of Fig. 14.55.

* **30.** Use Schematic Capture to calculate the output voltage in the circuit of Fig. 14.56.

* **31.** Use Schematic Capture to obtain the output waveform for a 2 V step input to an integrator circuit, as shown in Fig. 14.20, with values of R = 40 kΩ and C = 0.003 μF.

*Please Note: Asterisks indicate more difficult problems.